Domestic vs. Foreign Law: Portfolio Dynamics of Sovereign Debt*

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Abstract

Emerging-market governments borrow both under their own law and under foreign law. If the government defaults, the governing law determines whether domestic or foreign courts settle disputes, and so how much investors recover. For twelve governments, we document that all borrow under both laws, that each issues its foreign-law debt at longer duration, and that issuing under foreign law costs a fee proportional to the money raised and independent of maturity. The yield gap between a government's bonds under the two laws widens with the probability of default. We build a quantitative model of sovereign default in which the government can commit neither to repay nor to limit its future borrowing. Through the lens of the model, the widening gap implies that investors expect to recover more on foreign-law debt. The government still borrows under domestic law because shorter debt is less exposed to dilution. The fee keeps it from issuing shorter foreign-law debt, which would recover more and be less exposed to dilution. Calibrated to Colombia, the model implies recoveries of 47 percent under foreign law and 40 percent under domestic law. Removing the fee raises welfare by 12 percent of the gain the model generates when aggregate risk and default costs are eliminated, mostly through changes in how the government borrows and defaults.

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1 Introduction

When a government defaults, its creditors cannot seize assets within its borders, and what they can seize abroad generally falls short of what they are owed. What they recover is instead negotiated with the government, which remains excluded from financial markets until the two sides settle (Eaton and Gersovitz, 1981; Aguiar and Gopinath, 2006; Arellano, 2008; Yue, 2010). The bonds' terms set the rules of that negotiation; for example, they may set the majority of bondholders that must accept a deal for it to bind all of them. Whether the government can change those terms depends on the bonds' governing law, the law under which the terms are written and enforced. Under domestic law, the government can change them as it would any other law, even after the bonds have been sold. Under foreign law, a court outside the government's reach enforces them as written at issuance. In its 2012 restructuring, Greece used this difference to gain an advantage in the negotiation: it changed the terms of its Greek-law bonds but could not do so for its foreign-law bonds, and foreign-law bondholders recovered more on average (Zettelmeyer et al., 2013).

This paper studies how governments borrow under domestic and foreign law, empirically and quantitatively. We ask why governments borrow under both laws, and why they borrow longer under foreign law. The answer starts from how much investors expect to recover under each law. Estimates from past restructurings vary widely across countries and episodes (Cruces and Trebesch, 2013), and say little about what investors expect from a government that has not defaulted recently. We therefore develop a methodology that infers the expected recovery under each law from a country's own bond prices and the amount of debt it issues under each law, before any default occurs. We apply it to Colombia and find that investors expect to recover 40 percent of their claims under domestic law and 47 percent under foreign law. We then explain why Colombia issues under both laws, and measure what it would gain if its domestic-law debt offered investors the same protection in default as its foreign-law debt.

We begin by documenting how emerging-market governments use the two laws. We combine debt-office records with bond-level data from Dealogic on every bond twelve governments issued between 2006 and 2024, and with daily bond prices. Four facts emerge. First, all twelve governments issue under both laws, with foreign-law shares of marketable debt between 5 percent in Brazil and 82 percent in Paraguay. Second, foreign-law debt is issued at longer duration in every country. Third, issuing under foreign law costs a fee proportional to the money raised, whatever the maturity of the bond. Fourth,

the yield gap between the two laws widens with default risk. We measure it on the bonds that Brazil, Colombia, Paraguay and Russia issue in their own currency under foreign law. Following Chamon et al. (2018), we price each bond's payments on the domestic-law yield curve, so that the two bonds differ only in their governing law. In Colombia, the gap rises by 23 basis points for every 100-basis-point rise in the government's credit default swap (CDS) spread.

We then develop a quantitative model of sovereign default in which a government that lacks commitment chooses how much to borrow under each law (Arellano, 2008; Chatterjee and Eyigungor, 2012). Debt under each law has its own recovery and its own duration. Calibrated to Colombia, with durations of 3.75 years under domestic law and 8.71 years under foreign law, the model needs two moments to pin down the two recoveries: the slope of the yield gap and the share of debt under foreign law.

The model explains why a government that lacks commitment issues under both laws. The government can commit neither to repay nor to a path of future borrowing. On the one hand, foreign-law debt recovers more, which protects investors against default. On the other hand, domestic-law debt is shorter, which protects them against dilution: new borrowing lowers the value of debt that is still outstanding, and long debt stays exposed for longer (Hatchondo et al., 2016; Aguiar et al., 2019). Each law addresses one of the two commitment problems, so the government uses both. With equal durations, it issues only the high-recovery bond. With equal recoveries at the observed durations, it issues only the shorter bond.

Next, we ask why governments do not issue short foreign-law bonds, which would combine the higher recovery of foreign law with protection against dilution. Foreign-law bonds are placed through bank syndicates for a fee, while domestic-law bonds are auctioned for free in six of the ten countries where we observe the cost and for at most 1.5 basis points in the other four (Section 2.4). The fee is a tax on rolling over debt, so it weighs most on short bonds (Section 6.1). With only one-period bonds, a fee the size of the one we measure leaves most of the government's debt under foreign law. With a long foreign-law bond available, the same fee keeps the government from issuing short foreign-law debt, because it can offer investors the recovery of foreign law through the long bond, on which it pays the fee less often (Section 6.2). In the model calibrated to Colombia, a fee of 25 basis points cuts short foreign-law debt from 36 to 3 percent of the government's debt.

Last, we use the model to measure what the fee costs the government. Without the fee,

the government would issue no domestic-law debt, and its short debt would be issued under foreign law (Section 6.2). We give the short bond the recovery of foreign-law debt and keep its duration. The government borrows more and defaults less often. Spreads fall on both bonds, and more on the short bond, which now recovers more. Empirical work values foreign law by the premium on foreign-law bonds (Chamon et al., 2018; Saiegh and Biglaiser, 2024). In our model, most of the welfare gain comes from changes in how the government borrows and defaults. Political frictions make governments more impatient than their households (Cuadra and Sapriza, 2008; Aguiar and Amador, 2011). As in Aguiar et al. (2020), we evaluate the gain for households more patient than the government, and they also gain. In the model calibrated to Colombia, welfare rises by 8.2 basis points of permanent consumption at zero debt and by 9.0 at the mean level of debt, and 7.4 of the 8.2 come from the government's response.

Related literature. Chamon et al. (2018) show that in the euro area the yield gap between foreign-law and domestic-law bonds rises with default risk. Most foreign-law bonds are issued in a different currency from domestic-law bonds, so their main estimates convert currencies. In emerging markets, defaults come with large depreciations, which prevents us from following this approach. Saiegh and Biglaiser (2024) use identical dollar bonds that Argentina issued under both laws in 2020, and find that the foreign-law bonds trade at higher prices. Such twins are rare. We compare local-currency bonds issued under both laws, which several emerging markets have. We study four of them and give the first estimates for Brazil, Colombia and Paraguay. In all four, the gap rises with default risk.

Estimates of how much creditors recover in default come from past restructurings, mostly of external debt (Cruces and Trebesch, 2013). These episodes are rare for any one government, and their recoveries vary widely. Domestic law governs between 65 and 95 percent of the marketable debt in ten of the twelve governments we study, and evidence on defaults on this debt is recent (Erce et al., 2022). Arellano et al. (2023) and Arellano and Barreto (2025) estimate recoveries by matching a quantitative model of default to the data. They treat the recovery on each type of debt as a fixed parameter, which summarizes the outcome of the negotiation between the government and its creditors. We follow this approach for the two laws, taking the durations from the data and matching the share of debt under each law and how the yield gap moves with default risk. (See also Sturzenegger and Zettelmeyer 2008 and Graf von Luckner et al. 2025 on haircuts; Yue 2010, Benjamin and Wright 2009 and Asonuma and Trebesch 2016 on negotiation; and Schumacher et al. 2018 and Fang et al. 2021 on litigation and holdouts under foreign

law.)

We extend the quantitative models of sovereign default of Aguiar and Gopinath (2006) and Arellano (2008) by letting the government choose the law under which it borrows. As in Chatterjee and Eyigungor (2012) and Hatchondo et al. (2016), the government cannot commit to its future borrowing, so new debt dilutes the value of long-term debt. Several papers let the government choose among types of debt that serve different purposes. In Arellano and Ramanarayanan (2012), short-term debt gives incentives to repay, and long-term debt hedges against changes in spreads. In Ottonello and Perez (2019), foreign-currency debt avoids the costs of inflation, and local-currency debt provides a hedge. Choi et al. (2026) let the government choose currency and maturity together, and Arellano and Barreto (2025) let it default selectively on official and private lenders. In our model, the same investors hold both bonds, and the government defaults on both at once. The two bonds differ in their governing law, which sets how much investors recover, and in their duration. We take the durations from the data. (See also Eaton and Gersovitz 1981, Hatchondo and Martinez 2009, Du et al. 2020 and Passadore et al. 2026. As in Passadore and Xu 2022, a government that returns to the market still carries part of its defaulted debt; Yue 2010 and Mihalache 2020 model how that debt is renegotiated.)

We also contribute to work on the maturity of sovereign debt. A government that cannot commit to its future borrowing has a reason to borrow short, because short-term debt is less exposed to dilution (Hatchondo et al., 2016). A cost of issuing works the other way: long debt is rolled over less often, so it pays the cost less often (Jungherr et al., 2024). In Bigio et al. (2022), the cost comes from auctions in which larger issues fetch lower prices, and the government spreads its issuance across maturities. The cost we study is the fee that banks charge to place foreign-law bonds. This fee keeps the government from issuing short foreign-law bonds. In a simple model without default, we show that with a fee proportional to the money raised the government issues only the longest bond (Section 6.1). We measure what the fee costs the government using the welfare decomposition of Aguiar et al. (2020). (See also Missale and Blanchard 1994, Jeanne 2009, Broner et al. 2013 and Aguiar et al. 2019 on maturity; and Lee et al. 1996, Altınkılıç and Hansen 2000 and Melnik and Nissim 2003 on underwriting fees.)

2 How Governments Use the Two Laws

Every government we study borrows under both laws, issues its foreign-law debt at longer duration, and pays a fee to issue under foreign law that is a fixed share of the

money it raises.

2.1 Data

Our sample covers the central governments of twelve emerging markets from 2006 to 2024: Brazil, Chile, Colombia, Hungary, Mexico, Paraguay, Peru, Poland, Russia, South Africa, Turkey and Uruguay. We study their marketable debt: central-government bonds that any private investor can buy and later resell, net of the central bank's holdings. This definition excludes four kinds of public debt. The first is loans from multilateral and bilateral lenders. The second is bills issued by the central bank and the debt of state companies. The third is retail savings bonds. The fourth is bonds sold directly to public institutions, such as state pension funds or development banks, without an auction. These institutions are often required by law to hold government debt, a form of financial repression (Reinhart and Sbrancia, 2015; Chari et al., 2020), so their purchases do not reflect the price at which private investors would lend. Bonds that public institutions buy at the government's regular auctions remain in the sample.

Domestic-law bonds come from each debt office's records, which report for every placement its date, amount, coupon, maturity and yield at issuance. Foreign-law bonds come from Dealogic's bond-level issuance data, which report the same variables together with the governing law. On both sides we keep only bonds the central government itself issues, and we treat each re-tap of an outstanding bond as a separate placement. Daily bond prices come from Bloomberg, official sources and Capital IQ Pro. When more than one source covers a bond, we use the source with the widest coverage and check the results against the others. Daily five-year CDS spreads come from Markit, through Capital IQ Pro. Underwriting fees come from Dealogic, and the issuer's own expenses from the final prospectuses that governments file with the SEC. GDP in current US dollars comes from the World Bank's World Development Indicators.

2.2 Every government borrows under both laws

Every government in our sample borrows under both laws. Table 1 reports each government's marketable debt under each law, as a percentage of GDP and as a share of its total marketable debt, averaged over 2006–2024. Paraguay enters from 2013, the year of its first foreign-law bond. Foreign-law bonds make up between 5 percent of marketable debt in Brazil and 82 percent in Paraguay, and between 20 and 35 percent in seven of the twelve countries. As a percentage of GDP, foreign-law debt ranges from 1.5 percent in Russia to

Table 1. Marketable Debt by Governing Law

	Domestic law		Foreign law		Total
	% GDP	% debt	% GDP	% debt	% GDP
	(1)	(2)	(3)	(4)	(5)
Brazil	41.9	95.0	2.2	5.0	44.1
Chile	14.5	76.2	4.7	23.8	19.2
Colombia	14.3	70.3	6.1	29.7	20.4
Hungary	33.6	69.8	14.7	30.2	48.3
Mexico	23.5	84.0	4.5	16.0	28.0
Paraguay	1.4	17.8	8.9	82.2	10.3
Peru	10.2	65.3	5.2	34.7	15.4
Poland	29.6	79.2	7.8	20.8	37.4
Russia	6.5	79.1	1.5	20.9	8.0
South Africa	41.4	91.5	3.9	8.5	45.3
Turkey	19.2	75.5	5.5	24.5	24.7
Uruguay	11.8	37.5	20.1	62.5	31.9

Notes: Marketable central-government debt outstanding at each 31 December, averaged over 2006–2024, except Paraguay, averaged over 2013–2024, from its first foreign-law bond. Marketable debt consists of the central-government bonds that any private investor can buy on the same terms and later resell, net of the central bank's holdings. Columns (1) and (3) report the domestic-law and foreign-law stocks as a percentage of GDP, columns (2) and (4) as a percentage of total marketable debt, and column (5) the total as a percentage of GDP. Debt and GDP are in current US dollars, GDP from the World Bank (NY.GDP.MKTP.CD). Domestic-law stocks come from each debt office's records and foreign-law stocks from Dealogic. Robustness checks on column (4) are in Table 12.

20 percent in Uruguay, and total marketable debt from 8 percent in Russia to 48 percent in Hungary.

We measure the stock of marketable debt under each law at the end of each year from 2006 to 2024. A bond counts as outstanding from its settlement date until it matures. The domestic-law stock comes from the debt offices' records, with inflation-indexed bonds valued at what the government owes on them, and we convert it to dollars at the year's average exchange rate. The foreign-law stock is the dollar value of each bond at issuance, from Dealogic. Appendix A.1 reports the shares under other exchange-rate conventions, with foreign-law bonds revalued each year, and without the holdings of all public institutions.

2.3 Foreign-law debt is issued at longer duration

Table 2. Duration at Issuance by Governing Law

	Domestic law				Foreign law				Gap	
	Median (1)	Mean (2)	Face (3)	GDP (4)	Median (5)	Mean (6)	Face (7)	GDP (8)	Median (9)	Face (10)
Brazil	3.47	4.35	2.34	2.28	8.40	9.65	9.87	9.82	4.93	7.53
Chile	8.85	10.31	8.36	8.57	9.90	12.13	12.35	12.26	1.05	3.99
Colombia	3.75	4.74	5.79	5.52	8.71	10.56	10.58	10.62	4.96	4.79
Hungary	2.84	3.74	2.46	2.35	7.54	7.60	8.45	8.39	4.70	5.99
Mexico	0.48	2.67	2.00	1.96	8.87	10.14	11.12	11.13	8.39	9.12
Paraguay	3.46	3.88	4.08	3.87	8.74	10.54	10.51	10.54	5.28	6.43
Peru	8.64	9.28	9.10	9.30	11.77	13.46	13.26	12.97	3.13	4.16
Poland	2.56	3.97	3.72	3.63	8.19	8.75	8.80	8.80	5.63	5.08
Russia	5.80	5.49	4.54	4.64	8.00	8.98	9.51	9.55	2.20	4.97
South Africa	6.88	6.89	3.57	3.59	8.79	10.12	10.23	10.16	1.91	6.66
Turkey	1.43	2.25	2.13	1.86	7.64	7.83	7.43	7.57	6.21	5.30
Uruguay	8.19	8.55	8.91	8.78	9.45	10.55	11.51	11.47	1.26	2.60

Notes: Macaulay duration of each placement at the moment of sale, in years. Median and Mean count every placement once; Face weights each placement by the dollars it raised; GDP weights it by those dollars over its own year's GDP. Gap is foreign law minus domestic law. Samples start in 2004 or 2006 and end in 2024. Maturity at issuance, a common-window check and the governing law of foreign-law placements are in Tables 14, 15 and 16.

Foreign-law debt is issued at longer duration than domestic-law debt in every country in our sample. Table 2 reports the duration at issuance of each government's placements under each law. The median foreign-law placement has a duration between 7.5 years in Hungary and 11.8 years in Peru, and the median domestic-law placement between 0.5 years in Mexico and 8.9 years in Chile. The gap between the two medians ranges

from 1.1 years in Chile to 8.4 years in Mexico. Figure 1 plots the distribution of duration at issuance under each law. In most countries, domestic-law placements concentrate at short durations, where foreign-law placements are rare.

We measure the duration of every placement of marketable debt at the time of sale, following Macaulay (1938): the average of the years until each of the bond's payments, each weighted by its discounted value relative to the bond's price. For a bond with face value of 1 that pays coupon c at $n_1, \ldots, n_J$ years after settlement and is sold at yield y,

$$d = \frac{1}{p}\left[\sum_{j=1}^{J} \frac{n_j\, c}{(1+y)^{n_j}} + \frac{n_J}{(1+y)^{n_J}}\right], \qquad p = \sum_{j=1}^{J} \frac{c}{(1+y)^{n_j}} + \frac{1}{(1+y)^{n_J}}.$$

As in Arellano and Ramanarayanan (2012), we use duration because it is comparable across bonds with different coupons. A re-tap of an outstanding bond counts as a separate placement, measured from its own settlement date. For each law we report the median and mean across placements, and two weighted means: one weights each placement by the dollars it raised, and the other by those dollars over its year's GDP. Appendix A.3 describes how we treat each type of bond, and reports maturity at issuance, the medians over a common 2006–2024 window, and the laws under which the foreign-law placements are written.

2.4 Issuing under foreign law costs a fee proportional to the money raised

Issuing under foreign law costs the government a fee proportional to the money it raises, at a rate that does not depend on the maturity of the bonds or on the amount issued. A government that sells a foreign-law bond hires a syndicate of banks to place it with investors, and the banks keep a share of the proceeds, the underwriting fee. Domestic-law bonds are sold at auctions run by the central bank or the debt office. Because the fee is paid each time a bond is issued, a government that borrows short under foreign law pays it more often for the same debt. Section 6.2 uses this fact to explain why governments issue foreign-law bonds at longer duration, as we document in Section 2.3.

Dealogic reports the underwriting fee of each bond as a share of its face value. On average the bonds sell at par: the mean price, weighted by face value, is 99.9 percent of face value. A share of face value is therefore also a share of the money raised. For each bond, we divide the fee paid in dollars by the money raised. We restrict the sample to bonds issued under New York law, for which Dealogic records a fee for 92 percent of the bonds, against

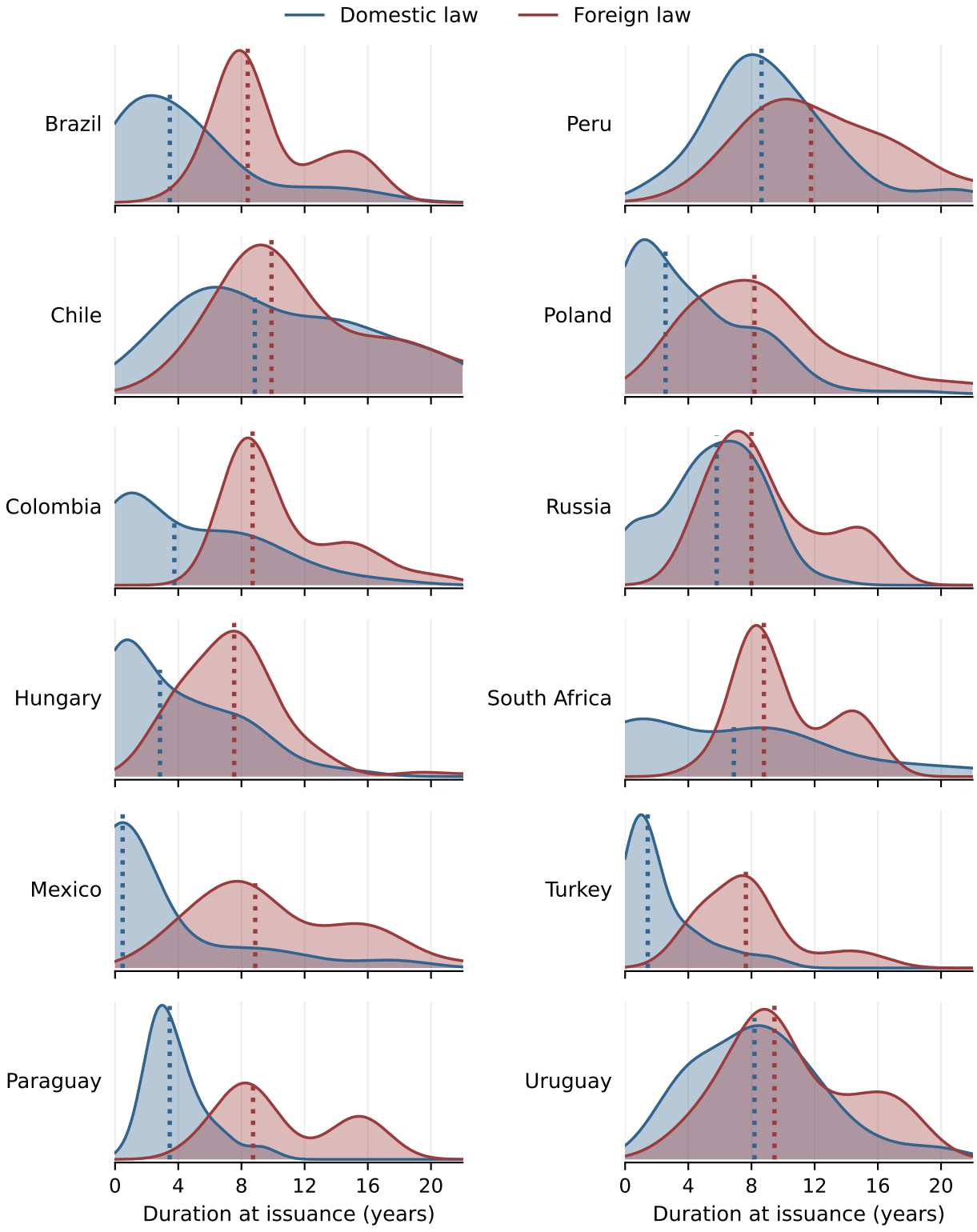


Figure 1. Distribution of Duration at Issuance by Governing Law

Notes: Kernel densities of duration at issuance, in years, over each government's placements under each law, 2006–2024 (2004–2024 for Hungary, Poland, Russia, South Africa and Turkey). Each placement, including each re-opening, counts once. Dotted lines are the medians of Table 2. The horizontal axis stops at 22 years; 0.8 percent of placements lie beyond it and enter the estimates.

9 percent of English-law bonds and 43 percent of Japanese-law bonds. The sample has 367 bonds issued by ten governments between 2006 and 2024; Dealogic records no fee for Paraguay and Russia. Appendix A.15 repeats the analysis for the bonds issued under other foreign laws: bonds issued under English law in euros pay 14 basis points, against 15 for New York-law bonds, while bonds issued in yen and Swiss francs pay more.

Table 3. Underwriting Fee on Foreign-Law Bonds

	Median fee	Mean fee
Brazil	23.6	24.0
Chile	4.0	5.6
Colombia	25.1	27.0
Hungary	18.2	19.0
Mexico	18.6	19.7
Peru	4.2	7.8
Poland	14.1	14.9
South Africa	13.2	13.8
Turkey	8.0	8.2
Uruguay	10.0	11.4

Notes: Underwriting fee in basis points of the money raised, New York-law bonds, 2006–2024.

Each government pays its own fee as a share of the money it raises, and the share does not depend on the maturity or the size of the bond. The median fee ranges from 4 basis points in Chile and Peru to 24 in Brazil and 25 in Colombia (Table 3). After removing country and year averages, the fee does not vary with maturity or with size (Figure 2).

For bond i, issued by government c in year t, we estimate

$$\text{fee}_{ict} = \beta_M \, \text{maturity}_{ict} + \beta_S \, \text{size}_{ict} + \alpha_c + \gamma_t + \varepsilon_{ict}, \tag{1}$$

where fee is the underwriting fee in basis points of the money raised, maturity is the years to maturity at issuance, and size is the money raised in US$ billion. The country effects α_c absorb the rate each government pays, and the year effects γ_t absorb market conditions in the year of issue. We cluster standard errors by country. With ten clusters, we also report wild-bootstrap p-values. Neither coefficient is significant, and we cannot reject that both are zero ($p = 0.23$; Table 4, Figure 2, Appendix A.15). The fee varies across governments and years. Without the country and year effects, maturity and size explain 2 percent of the variance of the fee, and neither coefficient is significant (column 1). With them, the R^2 rises to 0.87, and each set of effects is jointly significant ($p < 0.001$). Earlier evidence on

government bonds points the same way. Among the 37 government and agency bonds placed in 1996–97 that Melnik and Nissim (2003) study, the fee is unrelated to maturity and to size. Among the emerging-market government bonds placed between 1993 and 2007, the fee does not rise with the spread at which the bond is sold (Flandreau et al., 2010), so it does not price the risk of each bond either.

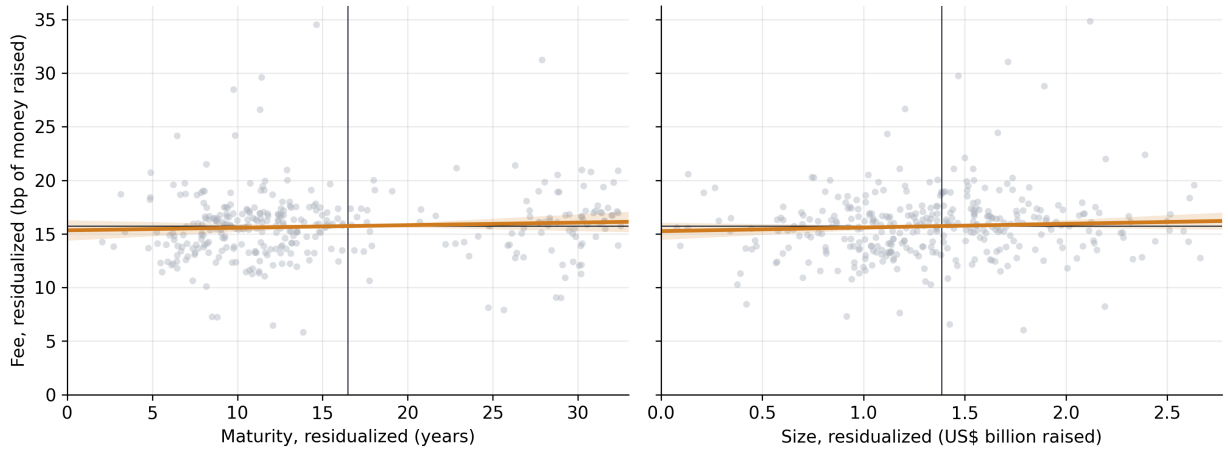


Figure 2. Underwriting Fee, Maturity and Size

Notes: Underwriting fee in basis points of the money raised, against maturity at issuance (left) and money raised in US$ billion (right), 360 New York-law bonds, 2006–2024. Each variable is net of country and year effects and of the other regressor, plus its sample average (fee 15.7 basis points, maturity 16.5 years, size US$1.4 billion). Line: fit of equation (1), with its 95 percent interval. Bonds beyond twice the average on either axis, 17 for maturity and 9 for size, are in the fit but not drawn.

Table 4. Regression of the Underwriting Fee on Maturity and Size

	(1)	(2)
Maturity (years)	0.050	0.024
	(0.057)	(0.029)
	[0.50]	[0.51]
Size (US$ billion raised)	−1.448	0.342
	(1.740)	(0.294)
	[0.46]	[0.27]
Country effects	No	Yes
Year effects	No	Yes
Countries (clusters)	10	10
Bonds	360	360
R^2	0.019	0.873

Notes: Dependent variable: underwriting fee in basis points of the money raised. Bonds of Table 3 without the 1 percent shortest and longest (Appendix A.15). Standard errors clustered by country in parentheses; wild-bootstrap p-values in brackets.

The government also pays a flat amount for each deal, which is similar across countries and deals. These are its own expenses: lawyers, listing and paying agents, which the prospectuses filed with the SEC state as a single estimate. The median ranges from US$120,000 in Uruguay to US$305,000 in South Africa. A deal twice as large costs 14 percent more in expenses, and expenses account for between 4 and 20 percent of the cost of a deal (Table 24).

When the debt office runs the auctions, as in Hungary and Peru, the government pays no agent. When the central bank runs them, Brazil, Mexico, Russia and South Africa pay nothing. Chile, Colombia, Poland and Uruguay pay at most 1.5 basis points, and Chile less than 0.13.[1]

3 The Yield Gap and Default Risk

In the four countries where we estimate it, the yield gap between domestic- and foreign-law bonds widens with the probability of default. Our sample includes foreign-law bonds that pay in the government's own currency. For each of them we construct a domestic-law bond with the same coupons and maturity. If the government repays, the two bonds pay the same. If it defaults, how much investors recover depends on the governing law. If investors expect the same recovery under both laws, the gap between the yields does not move with the probability of default. If they expect to recover different amounts depending on the law, the gap widens as default becomes more likely, because default is the only state in which the two bonds pay differently.

3.1 Pricing a bond under each law

Consider two one-year bonds of the same government. Both pay one unit of local currency and differ only in their governing law. The government defaults with probability p, and it defaults on both bonds at once.[2] If the government defaults, the law-j bond pays $\nu_j < 1$ units of local currency instead of one. After a default, disputes over domestic-law bonds go to domestic courts, and disputes over foreign-law bonds go to foreign courts.

[1] Chile pays a fixed amount for each issuance programme, and Uruguay 1 basis point a year on the stock of domestic-law bonds. The figures for Colombia and Poland are upper bounds. Colombia's is all the central bank's commissions for managing government funds, over the domestic-law bonds it manages, in 2025. Poland's is everything the government spent in 2017 to issue domestic bonds, apart from retail savings bonds, over what its auctions raised.

[2] In most defaults in which foreign investors could hold domestic-law debt, the government restructured both classes of bonds (Appendix A.9).

We interpret ν_j as the value of what holders of the law-j bond expect to receive once the default is resolved. The same investors price both bonds. They are risk neutral in dollars and discount at the dollar interest rate r. The exchange rate is $\mathcal{E}$ today, in local currency per dollar. Next year it is $\mathcal{E}'_R$ if the government repays and $\mathcal{E}'_D$ if it defaults.

The bond pays $1/\mathcal{E}'_R$ dollars if the government repays and $\nu_j/\mathcal{E}'_D$ if it defaults. Investors pay the expected dollar payoff discounted at r. The bonds trade in local currency, so their price in local currency is:

$$q_j = \mathcal{E}\,\frac{1}{1+r}\left[\frac{1-p}{\mathcal{E}'_R} + \frac{p\,\nu_j}{\mathcal{E}'_D}\right].$$

If $p = 0$, the two bonds have the same price, whatever their governing law. If $p > 0$, the gap between their prices is the value of the extra recovery, collected only in default. As default becomes more likely, holding the exchange rates fixed, this gap changes by

$$\frac{\partial\,(q_F - q_D)}{\partial p} = \frac{\nu_F - \nu_D}{1+r}\,\frac{\mathcal{E}}{\mathcal{E}'_D}.$$

If $\nu_F = \nu_D$, the gap does not move with the probability of default. If the recoveries differ, it widens as default becomes more likely, at a rate set by $\nu_F - \nu_D$. How much it widens depends on $\mathcal{E}/\mathcal{E}'_D$, the dollar value in default of one unit of local currency, relative to today. If the currency keeps its value in default, $\mathcal{E}/\mathcal{E}'_D = 1$, and each unit of local currency recovered next year is worth $1/(1+r)$ units today. The more value the currency loses in default, the lower $\mathcal{E}/\mathcal{E}'_D$ and the less the recoveries are worth, whatever ν_j is. In the limit, the gap vanishes.

3.2 The yield gap and depreciation upon default

In the data, we measure the gap in yields. A government has several foreign-law bonds outstanding, with different coupons and maturities, and yields put them on a common annual scale. We define the yield gap as the domestic-law yield minus the foreign-law yield. The logic of Section 3.1 carries over to yields: the yield gap does not move with the probability of default if the recoveries are equal, and widens if they differ. Only the exchange-rate term is different. In the price gap, the exchange rate enters through $\mathcal{E}/\mathcal{E}'_D$. We split it into two terms:

$$\frac{\mathcal{E}}{\mathcal{E}'_D} = \frac{\mathcal{E}}{\mathcal{E}'_R}\,\frac{\mathcal{E}'_R}{\mathcal{E}'_D}.$$

The first reflects the depreciation expected upon repayment. The second reflects the depreciation upon default, $\delta \equiv \mathcal{E}'_D/\mathcal{E}'_R$. As default becomes more likely, holding the ex-

change rates fixed, the yield gap changes by

$$\frac{\partial (y_D - y_F)}{\partial p} \simeq \frac{\nu_F - \nu_D}{\delta}$$

(Appendix A.5). The expected depreciation upon repayment lowers both prices in the same proportion, so it cancels in the yield gap. The depreciation upon default lowers only the dollar value of the recoveries. When $\delta > 1$, the slope is therefore a lower bound on the recovery difference. Section 3.5 measures δ and adjusts the gap for it.

3.3 Measuring the yield gap

Six of our twelve governments have issued foreign-law bonds in their own currency: Brazil, Chile, Colombia, Paraguay, Russia and Uruguay. We study those of Brazil, Colombia, Paraguay and Russia (Table 5). Section 2.3 shows that foreign-law bonds are issued at longer duration than domestic-law bonds. To compare their yields, we construct a hypothetical domestic-law bond with the same coupons and maturity for each foreign-law bond, as in Chamon et al. (2018).

Table 5. Local-Currency Foreign-Law Bonds

Country	Law	Currency	Bonds	Instruments	Window
Brazil	New York	BRL	3	Global BRL 12.5% '22, 10.25% '28, 8.5% '24	2014–2020
Colombia	New York	COP	3	Global TES 9.85% '27, 7.75% '21, 4.375% '23	2017–2020
Paraguay	New York	PYG	3	Global 7.9% '31, 8.5% '35, 8.5% '38	2025–2026
Russia	English	RUB	1	7.85% '18 eurobond	2014–2017

Notes: All bonds are nominal, fixed-coupon and non-callable. Brazil's and Colombia's windows end at the COVID break, Russia's when its bond falls below one year to maturity, and Paraguay's at the end of available data.

The construction uses the domestic-law zero-coupon yield curve $z_t(\cdot)$ of each day. This curve must contain only domestic-law bonds. We use published curves where they do: Bloomberg's for Brazil and Colombia and the central bank's for Russia. For Paraguay, no such curve exists, so we fit a Nelson–Siegel curve each day to the domestic-law bonds quoted that day (Appendix A.10). We do not extrapolate the curve, so each foreign-law bond enters the sample once its remaining maturity falls within the curve's range.

A foreign-law bond has a daily price $q_{F,t}$ and outstanding payments $c_1, \ldots, c_n$ per unit of face on dates $s_1 < \cdots < s_n$, the last one including principal. The construction proceeds in

three steps. First, we compute its yield to maturity $y_{F,t}$ from its observed price,

$$q_{F,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + y_{F,t}\right)^{s_k - t}}. \tag{2}$$

Second, we discount its payments on the domestic-law curve,

$$\widetilde{q}_{D,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + z_t(s_k - t)\right)^{s_k - t}}. \tag{3}$$

Third, we compute the yield $\widetilde{y}_{D,t}$ that this price implies,

$$\widetilde{q}_{D,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + \widetilde{y}_{D,t}\right)^{s_k - t}}. \tag{4}$$

The yield gap is $\widetilde{y}_{D,t} - y_{F,t}$. The two bonds then share issuer, currency, coupons and maturity, and differ only in their governing law.

We measure the probability of default with the government's five-year credit default swap (CDS) spread. A CDS insures against a default by the government. The buyer pays an annual premium, the spread, while the government repays, and the seller covers the loss on the government's bonds if it defaults. The standard contract pays in dollars, and its spread rises with the probability of default. For each country, we estimate how the gap moves with it in first differences:

$$\Delta\left(\widetilde{y}_{D,i,t} - y_{F,i,t}\right) = \gamma\, \Delta \mathrm{CDS}_t + \Gamma' X_{i,t} + \mu_i + u_{i,t}, \tag{5}$$

where i indexes bonds and t trading days, and the μ_i are bond fixed effects. $X_{i,t}$ contains three controls: the change in the difference between the bid-ask spread of the foreign-law bond and that of the government's ten-year domestic-law bond, the change in the VIX, and the bond's time to maturity. The first captures changes in the relative liquidity of the two bonds. The second captures movements in the CDS spread that reflect global risk aversion. The third accounts for how the gap changes as the bond approaches maturity. Appendix A.12 also lets the effect of liquidity vary with the level of the CDS spread. We use first differences because we cannot reject a unit root in the gap or the CDS spread (Appendix A.11). Differencing also removes any constant wedge between the two bonds.

The coefficient γ measures the slope of Section 3.2 per unit of spread. In the model of Section 3.1, a positive γ implies that investors expect to recover more under foreign law. As default becomes more likely, they demand a larger premium to hold domestic-law

bonds. If $\delta > 1$, γ understates the difference in recoveries, and Section 3.5 adjusts it for the depreciation upon default.

3.4 The yield gap widens with the probability of default

Table 6. The Yield Gap and the CDS Spread

$\Delta(\widetilde{y}_{D,i,t} - y_{F,i,t})$	Brazil	Colombia	Paraguay	Russia
Δ CDS spread	0.694***	0.225***	0.169**	0.073***
	(0.102)	(0.059)	(0.078)	(0.026)
Governing law	New York	New York	New York	English
Controls	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	—
Bonds	3	3	3	1
Observations	3,332	1,347	713	633
R^2	0.098	0.036	0.043	0.078
Window	2014–2020	2017–2020	2025–2026	2014–2017

Notes: Estimates of γ in equation (5), country by country. $X_{i,t}$: changes in the bid-ask spread difference and in the VIX, and time to maturity (Section 3.3). Standard errors in parentheses are Driscoll–Kraay, robust to heteroskedasticity, autocorrelation and correlation across bonds. Russia has one bond, so its standard errors are Newey–West, robust to heteroskedasticity and autocorrelation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The gap widens with default risk in every country we study (Table 6). For every 100 basis point rise in the CDS spread, it widens by 69 basis points in Brazil, 23 in Colombia, 17 in Paraguay and 7 in Russia. Each estimate is significant at the 5 percent level or better. Appendix A.12 reports the estimates under alternative specifications, windows and samples of bonds.

Reading the coefficients as a difference in what the same foreign investors recover after a default event requires that these investors can hold both bonds and convert their payments into dollars on the same terms. Three conditions make this hold. First, these investors face no restriction on holding the domestic-law bond. Otherwise the gap would also price the cost of access. Each sample starts only after capital can enter and leave the country without controls or entry taxes. Over these windows nonresidents held between 15 and 26 percent of the domestic-law debt in Brazil, Colombia and Russia (Appendix A.6). Second, no tax falls on one bond and not the other. Only Colombia withholds tax from nonresidents on domestic-law bonds. Netting the tax moves its slope from 0.23 to 0.21 (Appendix A.7). Third, holders of both bonds convert coupons and principal at the same exchange rate. Most of the foreign-law bonds automatically credit investors'

accounts in dollars, converting each local-currency payment at the onshore rate when it is due. In all the countries we study, a holder of the domestic-law bond buys and sells the currency at that same rate (Appendix A.8).

We run a placebo on Peru's sol notes. Foreign investors can buy Peru's domestic-law bonds abroad through Global Depositary Notes. A depositary bank holds the bonds in custody in Peru and issues notes on them to offshore investors. Like the foreign-law bonds of Section 3.3, the notes are denominated in soles, pay investors in dollars and clear outside Peru. Their claim on the government remains under Peruvian law. We then reprice the notes on the onshore domestic-law yield curve. Both sides of the gap are therefore domestic-law claims, and by Section 3.1 the gap should not move with the probability of default. We apply the construction and the regression of Section 3.3 to three of these notes (Appendix A.14). The coefficient is -0.02, with a p-value of 0.56.

3.5 Adjusting for depreciation upon default

Currencies lose value when governments default. Mano (2013) studies 104 defaults between 1873 and 2008. In 88.5 percent of them, the local currency lost value against the dollar between the end of the year before the default and the end of the default year. On average it lost 17.6 percent of its dollar value.[3]

These figures compare two dates, so they also contain the depreciation that would have happened without a default. To measure the depreciation in default relative to repayment, Du and Schreger (2016), following Mano (2013), compare two CDS on the same government. In both, the buyer pays a spread while the government repays, and the seller covers the loss on the government's bonds if it defaults. The dollar CDS pays this loss in dollars. The local-currency CDS pays the same share of its notional in local currency. In default, the local currency is worth $1/\delta$ as many dollars as in repayment, so the local-currency spread is lower by the same factor. We repeat their analysis for the five governments in our sample with daily prices on both contracts since 2015: Colombia, Mexico, Russia, South Africa and Turkey (Appendix A.5). In default, the local currency is worth between 35 and 50 percent fewer dollars than in repayment, 39 percent on average (Table 7). In Colombia δ is 1.55: a dollar buys 55 percent more pesos in default than in repayment, so the peso is worth $1 - 1/1.55 = 35$ percent fewer dollars. For Mexico and Turkey the figures are 36 and 35 percent, against 36 and 37 percent in Du and Schreger (2016) for the years up to 2015.

[3]The loss is $1 - \mathcal{E}/\mathcal{E}'$, where $\mathcal{E}$ and $\mathcal{E}'$ are the units of local currency per dollar in December of the year before the default and in December of the default year, averaged over the 104 defaults.

Table 7. Depreciation upon Default

	δ	s.e.	Loss of dollar value (%)
Colombia	1.55	0.006	35
Mexico	1.55	0.007	36
Russia	1.61	0.004	38
South Africa	2.01	0.009	50
Turkey	1.54	0.005	35
Average	1.65		39

Notes: Mean of the daily ratio of the dollar to the local-currency five-year CDS spread, January 2015 to June 2025 (Mexico to December 2024, Russia to February 2022), with Newey–West standard errors. The loss of dollar value in default relative to repayment is $1 - 1/\delta$.

With these estimates, we multiply the daily gap in equation (5) by δ. The depreciation upon default lowers the dollar value of the recoveries, and multiplying by δ restores it. For Colombia and Russia, δ is the mean of their own daily ratios over each estimation window, 1.57 and 1.56. Brazil and Paraguay have no local-currency CDS over their windows, so we use the average of Table 7, 1.65. For every 100 basis point rise in the CDS spread, the adjusted gap widens by 115 basis points in Brazil, 35 in Colombia, 28 in Paraguay and 11 in Russia (Table 8). Over 2015–2025, the means of the daily ratios have standard errors below 0.01 (Table 7), so we treat δ as known, and each adjusted slope keeps the significance it has in Table 6. Across the values of δ in Table 7, the slope ranges from 107 to 139 basis points in Brazil and from 26 to 34 in Paraguay.

Table 8. The Yield Gap Adjusted for Depreciation upon Default

	Brazil	Colombia	Paraguay	Russia
Measured (Table 6)	0.69^{***}	0.23^{***}	0.17^{**}	0.07^{***}
δ	1.65^{a}	1.57	1.65^{a}	1.56
Adjusted	1.15^{***}	0.35^{***}	0.28^{**}	0.11^{***}
With δ from 1.54 to 2.01	[1.07, 1.39]		[0.26, 0.34]	

Notes: The adjusted slope is the measured slope times δ. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. ^{a}Average of Table 7.

3.6 Comparing bonds in different currencies

Our yield gap compares two bonds that pay in the government's own currency. In principle, the same comparison could use bonds that pay in a foreign currency, such as the dollar. In practice, few of the governments we study have domestic-law bonds in foreign

currency. Brazil, Chile, Mexico, Peru, Poland and South Africa have sold none in the market since 2006. Hungary, Paraguay, Russia, Turkey and Uruguay have had only a few. These are too few to fit the domestic-law curve that the method of Section 3.3 needs. The alternative is to compare twin bonds, which share currency, coupon and maturity and differ only in their law. Chamon et al. (2018) search the emerging-market government bonds issued since 1990 and find such pairs only in Argentina and Russia.

Another alternative compares the government's foreign-currency foreign-law bonds with its local-currency domestic-law bonds. This requires creating a synthetic domestic-law bond in foreign currency. The investor buys the domestic-law bond and enters a currency forward that turns its promised local-currency payments into foreign currency. The simple model of Appendix A.4 shows that the synthetic bond delivers the recovery difference only if the currency does not depreciate upon default. Chamon et al. (2018) apply it to euro-area governments and assume that a default by one of them would not depreciate the euro against the dollar. For the governments we study, the local currency is worth 35 to 50 percent fewer dollars in default than in repayment (Table 7). The forward is written on these promised payments, but in default the government pays only part of them. The investor buys the shortfall after the currency has lost value, for less than the forward priced it, and so gains exactly when the government defaults. The converted gap mixes this gain with any difference in recoveries between the two laws, so even the sign of its slope is uninformative about the difference in recoveries that investors expect (Appendix A.4).

The bonds we compare both pay local currency, so they require no forward. The depreciation upon default still lowers the dollar value of the recoveries, so the slope we estimate is a lower bound on the recovery difference (Section 3.2). Section 3.5 measures this depreciation and adjusts the slope for it.

4 Model

We build a quantitative model of sovereign default in which the government can commit neither to repay nor to limit its future borrowing. It borrows from risk-neutral investors under two laws. The two bonds differ in how much investors recover after a default and in their duration.

4.1 Environment

Time is discrete and denoted $t \in \{0, 1, 2, \dots\}$. A small open economy receives a stochastic stream of income y_t, which follows a first-order Markov process. The government is benevolent and maximizes the household's expected utility,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$ and u is strictly increasing and strictly concave.

The government borrows with two bonds, one under domestic law and one under foreign law. The model is real, so both bonds pay in units of the consumption good. Let b_j be the face value of law-j debt outstanding. Each period a fraction λ_j of it matures, as in Hatchondo and Martinez (2009) and Chatterjee and Eyigungor (2012), so b_D and b_F determine every future payment the government owes. With market access, the government repays the maturing principal and chooses the stocks it carries into next period, b'_D and b'_F. It sells new bonds at prices q_D and q_F, and consumption is

$$c = y - \sum_{j \in \{D,F\}} \lambda_j b_j + \sum_{j \in \{D,F\}} q_j \left[b'_j - (1 - \lambda_j) b_j \right]. \tag{6}$$

The second sum is the money raised under the two laws. The prices depend on income and on the stocks the government chooses, (y, b'_D, b'_F), and Section 4.4 derives them.

4.2 Timing

Figure 3 shows the sequence of events within a period. A government with market access enters the period owing b_D under domestic law and b_F under foreign law. It observes income y and decides whether to repay or default on both bonds. A government in default is excluded from credit markets. It loses $\phi(y)$ of output in each period of exclusion and consumes $y - \phi(y)$. Each period it regains access with probability θ.

Default does not erase the debt. In Arellano (2008) and Chatterjee and Eyigungor (2012) the government returns to the market with no debt. In our model, when it regains access, each defaulted unit of law-j bond is exchanged for $\nu_j < 1$ new bonds with the same duration. The government then owes $\nu_D b_D$ and $\nu_F b_F$ upon reentry. Defaulted bonds keep a value during exclusion. As in Passadore and Xu (2022), we track their prices, q_D^D and q_F^D.

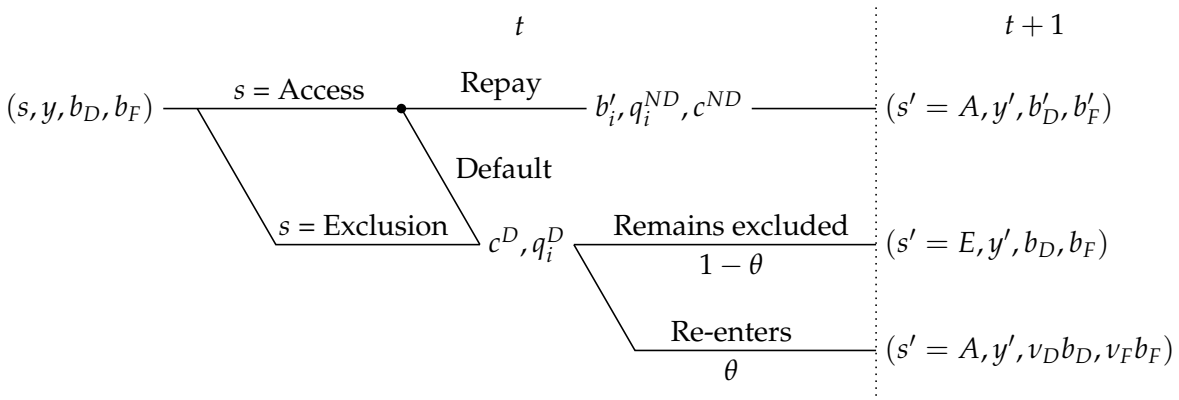


Figure 3. Timing

4.3 Recursive formulation

Upon repayment, the government takes the price schedules as given and chooses a vector of bond holdings for the next period (b'_D, b'_F). We look for a Markov equilibrium in which strategies are conditioned on (y, b_D, b_F). The unconditional value of the government is given by:

$$\widehat{V}(y, b_D, b_F) = \max_{d \in \{0,1\}} (1-d)\, V^R(y, b_D, b_F) + d\, V^D(y, b_D, b_F), \tag{7}$$

where d is the government's default policy:

$$d(y, b_D, b_F) = \mathbb{1}_{\left\{V^D(y,b_D,b_F) > V^R(y,b_D,b_F)\right\}}. \tag{8}$$

In case of default, there is no choice in the government's value:

$$\begin{aligned} V^D(y, b_D, b_F) &= u(c) + \beta \mathbb{E}_{y'|y} \left[(1-\theta) V^D\left(y', b_D, b_F\right) + \theta\, \widehat{V}\left(y', \nu_D b_D, \nu_F b_F\right) \right], \\ \text{s.t.} \quad & c = y - \phi(y). \end{aligned} \tag{9}$$

Under repayment, the government chooses next period's debt b'_D, b'_F:

$$\begin{aligned} V^R(y, b_D, b_F) &= \max_{b'_D, b'_F} u(c) + \beta \mathbb{E}_{y'|y} \left[\widehat{V}\left(y', b'_D, b'_F\right) \right] \\ \text{s.t.} \quad c + \sum_{j \in \{D,F\}} \lambda_j b_j &= y + \sum_{j \in \{D,F\}} q_j^{ND}\left(y, b'_D, b'_F\right) \left[b'_j - (1-\lambda_j) b_j \right] \end{aligned} \tag{10}$$

The choice of debt is also subject to an upper bound on the probability of default it implies, $\mathbb{E}_{y'|y}\left[d(y', b'_D, b'_F)\right] \leq \bar{p}$. Chatterjee and Eyigungor (2015) impose this bound and interpret it as an underwriting standard on new issues. Without it, an impatient gov-

ernment with positive recovery would borrow as much as it can at very low prices just before it defaults. The solution gives the policy functions for both bonds and, with them, the composition of the debt.

4.4 Bond prices

Competitive, risk-neutral investors buy both bonds and require the risk-free return r, as in Arellano (2008) and Chatterjee and Eyigungor (2012). A law-j bond issued by a government with market access is worth

$$q_j^{ND}\left(y, b'_D, b'_F\right) = \frac{1}{1+r}\,\mathbb{E}_{y'|y}\Big\{\left[1-d'\right]\left[\lambda_j + (1-\lambda_j)\,q_j^{ND}\left(y', b''_D, b''_F\right)\right] + d'\,q_j^{D}\left(y', b'_D, b'_F\right)\Big\}, \tag{11}$$

for $j \in \{D, F\}$, where $d' = d(y', b'_D, b'_F)$ and b''_j is the debt the government chooses next period. If the government repays, the holder receives the maturing fraction λ_j and values the remaining fraction $1-\lambda_j$ according to next period's price. That price depends on the government's next borrowing choice. If it defaults, the holder is left with a defaulted bond.

A defaulted law-j bond is worth

$$q_j^{D}\left(y, b_D, b_F\right) = \frac{1-\theta}{1+r}\,\mathbb{E}_{y'|y}\left[q_j^{D}\left(y', b_D, b_F\right)\right] + \theta\,\nu_j\,q_j^{ND}\left(y, \nu_D b_D, \nu_F b_F\right). \tag{12}$$

With probability $1-\theta$ the government stays excluded next period and the bond is then worth $q_j^D(y', b_D, b_F)$. With probability θ the government regains access next period. Before it does, each defaulted law-j bond is exchanged for ν_j new bonds with the same duration. The government enters next period owing $\nu_D b_D$ and $\nu_F b_F$ and can default on this debt again. Because of this, during the exchange the new bonds are valued at $q_j^{ND}(y, \nu_D b_D, \nu_F b_F)$.

The government defaults on both bonds at once and reenters on all its debt at once. When the two bonds have the same duration they differ only in what investors receive after a reentry.

Proposition 1. *Suppose $\lambda_D = \lambda_F$ and $\nu_F \geq \nu_D$. Then for any default and borrowing policies the prices in (11) and (12) satisfy in every state*

$$q_F^{ND}(y, b_D, b_F) \geq q_D^{ND}(y, b_D, b_F) \quad \textit{and} \quad q_F^{D}(y, b_D, b_F) \geq q_D^{D}(y, b_D, b_F).$$

If $\nu_F > \nu_D$ the inequalities are strict in every state from which the government later reenters and

then repays with positive probability. The first inequality holds with equality in every state from which the government never defaults.

Proof. Appendix A.19. □

4.5 Equilibrium

Definition 1. *A Markov Equilibrium with state variables* (y, b_D, b_F) *is given by:*

(i) A set of policy functions for consumption $c(y, b_D, b_F)$*, default* $d(y, b_D, b_F)$*, debt* $b'_D(y, b_D, b_F)$ *and* $b'_F(y, b_D, b_F)$*;*

(ii) A set of value functions $\widehat{V}(y, b_D, b_F)$*,* $V^D(y, b_D, b_F)$*,* $V^R(y, b_D, b_F)$*;*

(iii) Price schedules $q_D^D(y, b_D, b_F), q_F^D(y, b_D, b_F), q_D^{ND}(y, b'_D, b'_F), q_F^{ND}(y, b'_D, b'_F)$

such that:

1. *Given prices, the value functions solve: (7), (9), and (10), with associated policy functions;*
2. *Bond prices satisfy (11) and (12).*

4.6 Discussion of assumptions

We assume that the same risk-neutral foreign investors price both bonds. Nonresidents hold on average between 10 and 41 percent of the domestic-law debt of ten of our twelve governments (Table 18). This assumption allows us to abstract from the motive for selective default studied in the literature. We do not assume perfectly segmented markets as in Erce and Mallucci (2018), Di Casola and Sichlimiris (2025) and Shakhnov and Paczos (2026). In these models residents and nonresidents are assumed to hold different types of debt. This gives the government an incentive to default selectively.

We also assume that the government defaults on both classes of bonds at once. In the database of Erce et al. (2024), the government restructured both classes of bonds in 12 of the 14 episodes in which private foreign investors could hold bonds under both laws (Appendix A.9). In the model, a default excludes the government from both markets at a cost that does not depend on how much debt it defaults on. A government allowed to default selectively would therefore not choose to. It would bear the full cost of default and still have to service the other bond.

We also assume that the recoveries ν_D and ν_F are parameters that differ only by law. As in Arellano et al. (2023) and Arellano and Barreto (2025), the recovery is a parameter of

each debt contract. Governments renegotiate their defaulted debt with their creditors, and Yue (2010) and Benjamin and Wright (2009) model this negotiation. We do not model it. We estimate its outcome, the two recoveries, from the prices of the two bonds and the share of debt issued under each law (Section 5.1). Arellano and Barreto (2025) let the recovery differ between official and private debt. We let it differ between domestic-law and foreign-law debt.

We assume that the model is real, as in Arellano (2008) and Chatterjee and Eyigungor (2012). The model therefore studies outright default. It leaves out default through inflation, which lowers the real value of nominal domestic-law debt (Ottonello and Perez, 2019; Du et al., 2020; Choi et al., 2026). Indexed debt is protected from this channel, since its payments are fixed in real terms. In 2024, inflation-indexed bonds made up between 22 and 47 percent of the domestic-law debt of Brazil, Chile, Colombia, Mexico, South Africa and Uruguay (Appendix A.2). We return to this point in Section 5.1. There we discuss how the currency affects the yield-gap slope that the calibration targets.

Finally, we assume that the government issues one bond under each law, with a fixed duration. Each duration matches the median duration at issuance under that law (Section 5.1). The foreign-law bond is therefore the longer one, as it is for every government in our sample (Section 2.3). Sections 5.3 and 6 explain why a government that could issue a short and a long bond under each law would choose to issue only a long foreign-law bond and a short domestic-law bond.

5 Quantitative Analysis

Through the lens of the model, a yield gap that widens with default risk implies that investors recover more on foreign-law debt after a default. The government uses both laws because each protects investors against a different risk: with equal durations it issues only the bond with the higher recovery, and with equal recoveries at the observed durations only the shorter bond. Calibrated to Colombia, the model implies that investors expect to recover 47 percent of what they are owed under foreign law and 40 percent under domestic law.

5.1 Calibration

We calibrate the model to Colombia at an annual frequency. Colombia's debt office records every domestic-law placement, and Colombia files its foreign-law bonds with

the SEC. The shares, durations, yield gap and fee we target therefore all come from its own bonds. It is also a typical member of our sample. Its foreign-law share, 29 percent, lies within the range of seven of the twelve countries, and the slope of its yield gap lies between those of Russia and Brazil. Finally, Colombia has not defaulted since the 1930s. Without extrapolating from other countries' defaults, the recoveries its investors expect can only be inferred from prices, which is what our approach does. We solve a discretized version of the model with the algorithm of Gordon and Guerron-Quintana (2018). It extends the method of Chatterjee and Eyigungor (2012) to general choice sets, such as the choice between two bonds.[4]

We set seven parameters outside the model. Household utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$, with relative risk aversion $\sigma = 2$, the standard value in the literature. Log output follows an AR(1),

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \tag{13}$$

which we estimate on Colombia's annual real GDP from the World Development Indicators over 1976–2024, after removing a linear trend, as in Arellano (2008) and Chatterjee and Eyigungor (2012). This gives $\rho = 0.77$ and $\sigma_\varepsilon = 0.026$. The risk-free rate is $r = 1.8$ percent a year in real terms, following Bocola et al. (2019). The reentry probability is $\theta = 0.25$, a mean exclusion of four years. Estimates of the length of exclusion after a default range from three to six years (Gelos et al., 2011; Cruces and Trebesch, 2013). Coupons are normalized to zero, so each bond is a geometric-decay perpetuity whose Macaulay duration is $D = (1+r)/(\lambda+r)$. We set each λ so that this duration equals the median duration at issuance of Colombia's debt under that law (Section 2.3): 3.75 years for domestic law and 8.71 years for foreign law, which gives $\lambda_D = 0.254$ and $\lambda_F = 0.099$.

We choose the remaining five parameters jointly to match five moments of Colombia's data over 2006–2024: the discount factor β, the two recoveries ν_D and ν_F, and the two parameters of the output loss in default, which follows Chatterjee and Eyigungor (2012),

$$\phi(y) = \max\left\{0,\, d_0\, y + d_1\, y^2\right\}.$$

Three of them are the moments that quantitative models of sovereign default usually target (Arellano, 2008; Hatchondo and Martinez, 2009; Chatterjee and Eyigungor, 2012; Arellano and Ramanarayanan, 2012): the mean and the volatility of the spread on foreign-law bonds, 2.40 and 0.75 percent, from the J.P. Morgan EMBI Global Colombia, and the ratio of marketable debt to GDP, 20.9 percent (Section 2.2). In the model, debt is the face

[4] Our code builds on the replication files of Passadore and Xu (2022).

value of both bonds over output. The other two are new and discipline the recoveries: the foreign-law share of marketable debt, 29 percent (Section 2.2), and the slope of the yield gap on the CDS spread (Section 3.4). Colombia issues 98 percent of the face value of its foreign-law debt under New York law (Table 16), and the EMBI spread, the yield gap and the issuance fee of Section 2.4 all come from New York-law bonds, so we read the foreign-law bond in the model as a New York-law bond and its spread as the model counterpart of the EMBI spread.

We target how the yield gap moves with default risk and leave its average level untargeted. The level of the gap also reflects differences between the two bonds that do not depend on default risk, such as their liquidity and tax treatment (Section 3.4). The value of a higher recovery does depend on default risk, because investors collect it only if the government defaults. The slope of the gap on the CDS spread isolates this value, and differencing removes the wedges that stay constant.

To compute the slope in the model, we follow the construction of Section 3.3. We price five pairs of hypothetical bonds along the simulated equilibrium path, with the government's equilibrium default and borrowing policies. The two bonds in a pair differ only in their recovery. Each repays a fraction $\lambda_M = 1/M$ of its face value every year, so it matures in $M \in \{1, 3, 5, 7, 10\}$ years on average. Their durations span those of the bonds in Section 3. The yield gap for pair M is the difference between the spreads of its two bonds,

$$s^D_{M,t} - s^F_{M,t} = \lambda_M \left(\frac{1}{q^D_{M,t}} - \frac{1}{q^F_{M,t}} \right). \tag{14}$$

The model counterpart of the CDS spread is the spread on the bond with $M = 5$ and the foreign-law recovery, $s^F_{5,t}$. This matches how the contract settles. Under the standard ISDA terms for Latin American and for Emerging European and Middle Eastern sovereigns, only foreign-law bonds issued outside the domestic market can be delivered when the contract pays out (ISDA, 2017).[5]

The model slope γ is the estimate of

$$\Delta\left(s^D_{M,t} - s^F_{M,t}\right) = \gamma \, \Delta s^F_{5,t} + \alpha_M + \varepsilon_{M,t} \tag{15}$$

over the years with market access in a long simulation.[6] The maturity fixed effect α_M

[5] The terms are the same in the matrices of September 2014, December 2017, May 2022 and October 2025.

[6] We simulate 101,000 years and drop the first 1,000. The regressor is common to all maturities, so γ equals the average of the five slopes estimated maturity by maturity.

plays the role of the bond fixed effect, and holding each maturity constant plays the role of the maturity control. The model is real, so its slope has no currency effect. We target the measured slope. Section 5.2 adjusts it for depreciation upon default.

5.2 The calibrated model

Table 9. Parameters and Targets

A. Externally set				
Parameter	Value	Description	Source	
σ	2	Relative risk aversion	Standard	
ρ	0.77	Autocorrelation of log output	WDI, 1976–2024	
σ_ε	0.026	Std. deviation of output shocks	WDI, 1976–2024	
r	0.018	Risk-free rate	Bocola et al. (2019)	
θ	0.25	Reentry probability	Four-year exclusion	
λ_D	0.254	Share maturing each year, domestic-law debt	Duration, 3.75 years	
λ_F	0.099	Share maturing each year, foreign-law debt	Duration, 8.71 years	
B. Internally calibrated				
Parameter	Value	Main target	Data	Model
ν_D	0.40	Foreign-law share (%)	29.0	29.0
ν_F	0.469	Slope of the yield gap, γ	0.23	0.23
β	0.920	Marketable debt / GDP (%)	20.9	21.0
d_0	-0.188	Mean spread (%)	2.40	2.34
d_1	0.236	Spread volatility (%)	0.75	1.02

Notes: Annual frequency. Colombia, 2006–2024. The parameters in panel B are chosen jointly to match the five moments; each is listed next to the moment it affects most. The spread is the J.P. Morgan EMBI Global Colombia spread, and its model counterpart is the spread on the foreign-law bond.

Table 9 pairs each parameter with the moment it affects most, although all five are chosen jointly. The mean spread mainly disciplines d_0, its volatility d_1, and the debt ratio β. The two recoveries are pinned down jointly by the two new moments, as Figure 4 shows.

The slope of the yield gap on the CDS spread identifies the recovery gap $\nu_F - \nu_D$. The yield gaps in the model regression come from pairs of bonds with the same duration that differ only in their recovery (Section 5.1). Until the government defaults the two bonds of a pair have the same cash flows. After a default the foreign-law bond pays more when $\nu_F > \nu_D$. The larger the recovery gap, the more the yield gap responds to a change in the probability of default and so to the CDS spread. Panel (a) of Figure 4 shows the slope rising with $\nu_F - \nu_D$. Through the lens of the model the positive slopes of Section 3.4 imply that foreign-law debt recovers more in Brazil, Colombia, Paraguay and Russia. For

Colombia the slope of 0.23 implies a recovery gap of 7 percentage points.

Given the recovery gap, the foreign-law share of debt identifies the level of the two recoveries. Panel (b) holds the recovery gap at 7 percentage points and raises both recoveries together. Higher recoveries lower what investors lose in a default. The government then borrows more and in turn defaults more often. In the model the default probability rises from 3.0 to 4.9 percent as ν_F rises from 0.37 to 0.62. With more frequent defaults the larger cash flows of the foreign-law bond after a default matter in more states. This makes the foreign-law bond relatively more attractive to issue, and the foreign-law share rises with the level of recovery. Given a gap of 7 percentage points, the foreign-law share of 29 percent pins down recoveries of 40 percent on domestic-law debt and 47 percent on foreign-law debt.

The measured slope understates the recovery gap, because depreciation upon default lowers the dollar value of what investors recover on domestic-law debt (Section 3.5). Adjusted for this depreciation, Colombia's slope is 0.35. We hold the government's behavior at the calibration and widen the recovery gap of the bond pairs symmetrically, lowering the domestic-law recovery and raising the foreign-law recovery by the same amount. The model regression reproduces the adjusted slope at recoveries of 39 and 48 percent, a gap of 9 percentage points. We keep the baseline calibration in the rest of the paper. It does not rely on the estimates of δ, and its recovery gap of 7 percentage points serves as a lower bound on the difference between the two recoveries.

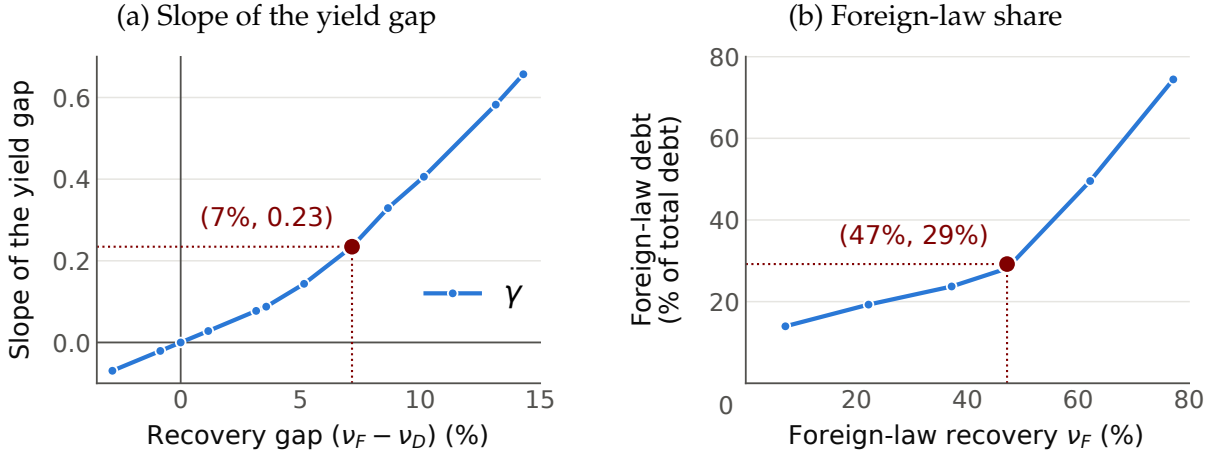


Figure 4. Identification of the Recoveries

Notes: Panel (a) plots the model slope of the yield gap on the CDS spread, γ, against the recovery gap $\nu_F - \nu_D$. Panel (b) plots the foreign-law share of debt against the foreign-law recovery ν_F, holding the recovery gap at its calibrated value. Other parameters at their calibrated values. The dots mark the calibration.

Table 10 reports untargeted moments. The model reproduces the volatility of consump-

Table 10. Untargeted Moments

	Model	Data
corr(spread, y)	-0.69	-0.12
corr(foreign-law share of issuance, y)	-0.69	-0.29
$\sigma(c)/\sigma(y)$	1.17	1.15
corr(c, y)	0.83	0.92
Debt service / GDP (%)	4.3	3.9
$\sigma(\log y)$	0.039	0.041
Default frequency (%)	3.7	—

Notes: Colombia, 2006–2024. Data moments for output and consumption use the cyclical component of logged series, HP filter with $\lambda = 100$, except $\sigma(\log y)$, which is implied by the linearly detrended AR(1); model moments use the simulated series. Model debt service includes amortization only, because coupons are zero; the data include coupons.

tion relative to output, 1.17 against 1.15 in the data. Debt service is 4.3 percent of GDP in the model and 3.9 percent in the data. Spreads and the foreign-law share of new issuance are more countercyclical in the model than in the data: their correlations with output are -0.69 in the model, against -0.12 and -0.29 in the data. As in other models in which output is the only shock (Arellano, 2008), default risk in the model rises only when output falls, so spreads move more closely with output than they do in the data. The model implies that Colombia defaults in 3.7 percent of years. The calibration infers this frequency from Colombia's spreads and debt, as it does the recoveries.

The model also generates the split between the two laws that we observe: the government holds 29 percent of its debt under foreign law and the rest under domestic law. Section 5.3 shows that this split requires the two bonds to differ in both recovery and duration.

5.3 The trade-off between recovery and duration

In the calibrated model the two bonds differ in their recovery and in their duration. Using counterfactual experiments that remove one difference at a time, we show that the government values foreign-law debt for its higher recovery and domestic-law debt for its shorter duration.

Both bonds carry the same default risk, since the government defaults on both at once. What differs is how much investors recover when it does. The foreign-law bond recovers more, so if the two bonds had the same duration it would trade at a higher price (Proposition 1). The government receives this higher price today but owes more when it reen-

ters (equation (12)). This later payment comes when the government owes little and the deadweight losses from default are gone. In the simulations its debt at reentry averages 10 percent of output, against an average of 21 percent, and new bonds sell at 97 percent of their risk-free price under domestic law and 89 percent under foreign law. Investors value the recovery at these prices and pay for it up front. As is standard in quantitative models of sovereign default, the government is more impatient than investors, with $\beta = 0.92$ against $1/(1+r) = 0.982$. Without this impatience the government would not borrow enough to default as often as in the data (Aguiar and Gopinath, 2006; Chatterjee and Eyigungor, 2012). The same impatience makes a higher recovery attractive, because it lets the government move consumption from the years after reentry to today. The higher price dominates. We give both bonds the same duration, between 1 and 8.71 years, keep the higher recovery on the foreign-law bond and solve the model again. At each duration we try the government issues only the foreign-law bond (Table 28).

The foreign-law bond is also longer. Duration has a benefit and a cost for the government. The benefit is insurance against bad times (Arellano and Ramanarayanan, 2012). In the model spreads rise when output falls. Next period a unit of law-j debt costs the government the share λ_j that matures plus the market value of the rest, $(1-\lambda_j)\, q_j^{ND}$. This value falls when output falls, which is when the government values resources the most. The longer the bond, the more of its value is owed at that market price. We measure this insurance as Arellano and Ramanarayanan (2012) do, with

$$H_j = \frac{\mathbb{E}\left[u'(c')\, R_j'\right]}{\mathbb{E}\left[u'(c')\right]\, \mathbb{E}\left[R_j'\right]}, \qquad R_j' = \lambda_j + (1-\lambda_j)\, q_j^{ND}, \tag{16}$$

where the expectations are over next period's states with repayment. The numerator weights the cost of the bond in each state by the government's marginal utility. The denominator is that cost if it did not move with marginal utility. A one-period bond costs one unit in every state, so its H_j equals one. A bond that is cheaper to owe in bad times has H_j below one.[7] The insurance measure $1 - H_j$ is 15.9 basis points for foreign-law debt and 11.3 for domestic-law debt, so on insurance alone the government would prefer foreign-law debt. This advantage comes from the longer duration. The higher recovery of foreign-law debt pulls the other way. Because investors lose less in default, the price of the bond falls less when default becomes more likely, and the bond is less cheap to owe in bad times. This pull is small. Giving the domestic-law bond the foreign-law recovery

[7] For example, take two equally likely states with marginal utility 1.5 and 0.5, and a bond that costs 0.70 in the bad state and 1 in the good one. Its average cost is 0.85, its cost weighted by marginal utility is 0.775, and $H = 0.775/0.85 = 0.91$.

lowers its measure from 11.3 to 10.1 basis points.

The cost of longer duration is dilution (Hatchondo et al., 2016; Chatterjee and Eyigungor, 2015). The price of a bond depends on the probability of default in every period until it is repaid. These probabilities depend on how much the government will borrow in the future, and the government cannot commit to its future borrowing. When it borrows again it does not account for the loss this imposes on bonds already outstanding (Hatchondo and Martinez, 2009). Investors anticipate that future borrowing will raise the probability of default and pay less for the bond today. In equation (11) this enters through next period's price, which depends on the government's next borrowing choice. A one-period bond is repaid next period, so its price depends only on the probability of default next period, which today's borrowing determines. A long bond is still outstanding when the government borrows again.

To see which of the two dominates at the observed durations, we give both bonds the same recovery, between 0 and 0.75, keep the calibrated durations of 3.75 and 8.71 years and solve the model again. Dilution outweighs insurance at every recovery: the long bond accounts for between 0.1 and 0.3 percent of debt (Table 30). The domestic-law bond is shorter, but it lasts more than one period. It also provides some insurance for the government, but it is less exposed to dilution than the long bond. The balance changes when the short bond lasts one year. A one-year bond is repaid before the government borrows again, so it is never diluted, and it provides no insurance. Against it the insurance of the long bond is enough for the government to hold between 48 and 68 percent of its debt in the long bond.

In the calibrated model the bond with the higher recovery is also the longer one. Neither bond then dominates the other. The foreign-law bond offers investors a higher recovery but more exposure to dilution. The domestic-law bond offers the reverse. The government therefore borrows under both laws and holds 29 percent of its debt under foreign law. A long domestic-law bond would lose on both counts and the government would not issue it. The bond that would win on both is a short foreign-law bond. Section 6.2 asks why governments do not issue it.

6 The Fee on Foreign-Law Debt

A fee proportional to the money raised weighs more on short debt, which must be issued again each time it matures, and keeps governments from issuing the short foreign-law bond of Section 5.3.

6.1 Fees and maturity

The fee on foreign-law bonds is a fixed share of the money raised, whatever the maturity of the bond. A government that borrows short must issue again each time its debt matures, so it pays the fee more often for the same debt. The fee therefore makes short foreign-law debt more costly than long foreign-law debt. Van der Wansem et al. (2019) show the arithmetic for a ten-year bond: about 30 basis points of fees and expenses paid upfront add 4.3 basis points to its annual cost. On a one-year bond the same costs are paid every year and add about 30 basis points. Appendix A.20 gives the general formula. In the model of Jungherr et al. (2024), issuance costs make short debt costly to roll over, which gives firms a reason to borrow long. Bigio et al. (2022) document that larger auctions of Spanish government bonds sell at lower prices, and show that such a cost leads the government to spread its issuance across maturities. The fee we document in Section 2.4 is a constant share of the money raised that does not depend on the maturity at issuance. As a first step, we study the fee in a simple model without the forces that the sovereign debt literature uses to explain maturity: default, dilution and hedging (Arellano and Ramanarayanan, 2012; Hatchondo et al., 2016; Aguiar et al., 2019). In this model the fee is the only force that shapes maturity, and the government issues only the longest bond (Proposition 2).

Consider a government that receives a constant endowment y and values consumption at $\sum_{t\geq 0}\beta^t u(c_t)$, with u strictly increasing and strictly concave. It borrows from lenders who discount at the interest rate r, and it is as patient as they are, $\beta = 1/(1+r)$. It always repays. It starts owing $B_0 > 0$, due at date 0. At date 0 it chooses one decay rate ξ for all the bonds it will ever issue. A unit pays $1, \xi, \xi^2, \ldots$ in the periods after it is issued, so ξ plays the role of $1-\lambda$ in Section 4. Its price is the value of these payments discounted at r:

$$q(\xi) = \sum_{k\geq 1} \beta^k \xi^{k-1} = \frac{\beta}{1-\beta\xi}.$$

The one-period bond is $\xi = 0$ and the consol is $\xi = 1$. Each issue or buyback pays a fee f on its market value.[8] Appendix A.21 gives the budget constraints and the debt limit.

Proposition 2 (Maturity under a fee)**.** *If $f = 0$, every maturity gives the same allocation. If $f \in (0,1)$ and $(1-f)\,u'(y-B_0) > u'(y)$, so that the government borrows at date 0, the consol is the unique optimal maturity: the government issues only at date 0 and never rolls its debt over.*

[8] Without a fee on buybacks, Proposition 2 holds except that the consol is no longer the only optimal maturity. Bonds whose payments grow, $\xi > 1$, are also optimal, because the government can buy back the growth in their payments at no cost.

Proof. Appendix A.21. □

The fee is paid every time the government issues and depends only on the money it raises. A government that refinances its initial debt does so once, with a consol issued at date 0. With shorter debt it would have to borrow periodically to keep its consumption constant and pay the fee each time.

6.2 Fees and the choice of law

Ten of the twelve governments sell domestic-law debt with durations under one year, and none sells foreign-law debt that short. In Colombia, the country to which we calibrate the model, the shortest foreign-law bond had a duration of 7.2 years at issuance, and the shortest domestic-law placement 0.2 years.

Section 6.1 shows that the fee on foreign-law debt weighs more on short bonds, because a short bond must be issued again each time it matures. We now use the quantitative model to ask whether at the fees we see in the data this effect is enough to keep governments from issuing short foreign-law bonds. Such bonds would protect investors against both default and dilution.

We add the fee to the model of Section 4. As a starting point, we consider a government that chooses between two one-period bonds, one under each law. It pays a fee f on the market value of the foreign-law bonds it issues or buys back, as in Section 6.1. The second experiment adds a long foreign-law bond, so we write the budget constraint for any number of bonds and collect the foreign-law bonds in the set $\mathcal{F}$:

$$c + \sum_j \lambda_j b_j \;=\; y + \sum_j q_j \left[b'_j - (1-\lambda_j) b_j \right] \;-\; f \sum_{j \in \mathcal{F}} q_j \left| b'_j - (1-\lambda_j) b_j \right|. \tag{17}$$

The pricing equations are those of Section 4, so the fee affects prices only through the government's default and borrowing policies. We set f at Colombia's median fee of 25 basis points (Table 3) and keep all other parameters at their values in Section 5.1. Without a fee the government borrows only under foreign law (Section 5.3). At 25 basis points it still holds 63 percent of its debt under foreign law (Table 31). Foreign-law debt falls to 1.5 percent of its debt at 75 basis points and to 0.04 percent at 100 basis points. That is three to four times Colombia's fee. With only one-period bonds, the government avoids the fee only by giving up the higher recovery.

When two bonds have the same duration, the foreign-law bond pays more only after a

default, through the new bonds investors receive when the government reenters (Proposition 1). The more likely a default, the more this later payment raises its price relative to the domestic-law bond. The fee takes the same share of the money raised whatever the probability of default. The fee needed to keep foreign-law debt below 1 percent of the government's debt rises with the probability of default, because the government gains more from issuing foreign-law debt when a default is more likely. It is 30 basis points when the probability of default in the next year is below 0.5 percent and 100 basis points when it is 1 percent or more. At 25 basis points, foreign-law debt is 1.5 percent of the government's debt in the first case and all of its debt in the second.

In the second experiment we add a long foreign-law bond. The government chooses among a short domestic-law bond, a short foreign-law bond and a long foreign-law bond. The two short bonds last one period, as before. The long bond has a duration of 2.32 years, so the ratio of the long to the short duration matches the one we see in Colombia's debt. The two foreign-law bonds share the recovery ν_F. They pay the same fee f, since in the data the fee does not depend on maturity (Section 2.4), in Colombia as in the other countries (Figure 7). All other parameters keep their values from Section 5.1.[9]

Without a fee, the government again issues no domestic-law debt. It holds 64 percent of its debt in the long foreign-law bond and 36 percent in the short one (Section 5.3). At Colombia's fee of 25 basis points, short foreign-law bonds fall to 2.9 percent of the debt (Table 31), and total debt stays at 19 percent of GDP. Keeping them below 1 percent of the debt takes 35 basis points, less than half the fee needed with only one-period bonds.[10] The long bond explains the difference. It carries the same recovery as the short foreign-law bond and pays the fee only on the share that matures each year. With a long foreign-law bond, the government can pay the fee less often without giving up the higher recovery.

As with one-period bonds, the fee needed to keep short foreign-law bonds below 1 percent of the government's debt rises with the probability of default. It is 20 basis points when the probability of default in the next year is below 0.5 percent and 40 basis points when it is 1 percent or more. At 25 basis points, short foreign-law bonds are below 0.1 percent of the government's debt in the first case and 4.8 percent in the second. The long bond lowers the fee needed most when default is likely: from 100 to 40 basis points, against 30 to 20 when the probability is below 0.5 percent. The higher recovery matters most when default is likely, and the long bond now delivers it.

[9] With three bonds the state has four variables: output and three debt stocks. We solve the model on a grid of 25 points for output and 40 for each stock.

[10] Without a fee on buybacks, the results do not change. The share of short foreign-law bonds moves by at most 0.3 percentage points at any fee.

Quantitatively, the fee we measure in the data is enough to keep the government from issuing short foreign-law bonds whenever it can also issue a long foreign-law bond and a domestic-law bond. Section 7 measures what the fee costs in terms of welfare.

7 Gains from Removing the Fee

Through the lens of our model, we analyze what the fee costs a government by keeping it from issuing short foreign-law bonds (Section 6.2). Without the fee, the government would issue no domestic-law debt, and its short debt would be issued under foreign law. In particular, we solve the model calibrated to Colombia for a specification in which the short bond takes the recovery of foreign-law debt, 47 percent instead of 40, and keeps its duration of 3.75 years. Unlike the experiments of Section 6.2, this exercise keeps the durations of the calibrated model, which match Colombia's debt. The rest of the parameters are set to the values of the baseline calibration. Welfare rises by 8.2 basis points of permanent consumption, 12 percent of the 71 basis points gained by eliminating aggregate risk and the costs of default in the baseline economy.

We measure welfare as the permanent percentage change in consumption Δ that makes the household indifferent between the baseline economy $\mathcal{E}_0$ and the economy without the fee $\mathcal{E}_1$,

$$(1+\Delta)^{1-\sigma}\,\bar{V}^{\mathcal{E}_0} = \bar{V}^{\mathcal{E}_1}, \qquad \bar{V}^{\mathcal{E}} \equiv \sum_{y} \pi(y)\, V^{\mathcal{E}}(y, b), \tag{18}$$

where π is the invariant distribution of output and $b = (b_D, b_F)$ is the debt at which we evaluate the gain. In $\mathcal{E}_1$ we solve the model again, so the government's borrowing and default policies adjust. We evaluate the gain at zero debt, as in Chatterjee and Eyigungor (2012) and Hatchondo et al. (2016), and at the mean level of debt in the baseline economy. It is 8.2 basis points at zero debt and 9.0 at the mean level of debt.

Table 11 shows how the economy changes when the short bond is issued under foreign law, relative to the baseline economy in which it is issued under domestic law. The government borrows more and defaults less often. Debt rises by 0.4 percentage points of GDP, and the default frequency falls by 0.3 percentage points a year. The lower default frequency lowers the spreads on both bonds. The spread on the long bond falls by 22 basis points. The spread on the short bond falls by 46 basis points, twice as much, because investors in the short bond also recover more after a default than in the baseline.

Removing the fee affects welfare through three forces. The first two act even when the government's borrowing and default policies stay fixed. First, investors expect to recover

Table 11. Removing the Fee on Foreign-Law Debt

A. Change in the economy	
Mean spread, short debt (bp)	−46
Mean spread, long debt (bp)	−22
Debt / GDP (percentage points)	+0.4
Default frequency (percentage points a year)	−0.3
B. Welfare gain (bp of permanent consumption)	
Total	+8.2
New prices, policies fixed	+2.1
New debt at reentry, policies fixed	−1.3
Both, policies fixed	+0.8
Change in borrowing and default policies	+7.4

Notes: Changes from the baseline economy to the economy without the fee, in which the short bond is issued under foreign law and recovers 0.47, model simulations. The welfare gain is Δ in equation (18), evaluated at zero debt. The change in policies is the total minus the gain with both changes and fixed policies. Appendix A.23 describes the decomposition.

more on short debt after a default, so they pay more for it. Second, the government reenters financial markets owing more. Third, the government changes how much it borrows and when it defaults. We measure the first two forces with the policies at their baseline values (Table 11, panel B). Giving the government the prices of the new economy, with the baseline debt at reentry, raises welfare by 2.1 basis points. Giving it the larger debt at reentry of the new economy, with baseline prices, lowers welfare by 1.3 basis points. With both changes and fixed policies, welfare rises by 0.8 basis points, so the remaining 7.4 basis points come from the government's response. Empirical work values foreign law by its price: foreign-law bonds trade at a premium, and more so as default risk rises (Chamon et al., 2018; Saiegh and Biglaiser, 2024). In our model the main driver of the gain is the change in the government's incentives to borrow and to default. Appendix A.23 describes these calculations.

So far we evaluate welfare with the government's discount factor, $\beta = 0.92$, below the investors', $1/(1+r) = 0.982$. This impatience relative to investors makes the higher recovery attractive, because it lets the government move consumption from the years after reentry to today (Section 5.3). Although the government maximizes the utility of a representative household, the two need not discount the future at the same rate. Political frictions make governments more impatient than their households (Cuadra and Sapriza,

2008; Aguiar and Amador, 2011). As in Aguiar et al. (2020), we therefore value the consumption that the government's policies generate with a household discount factor β_H. It ranges from 0.92, a household as impatient as the government, to 0.982, a household as patient as investors. The government keeps its discount factor, so its policies and the prices they imply do not change. We also follow them in decomposing the gain into three terms: default costs, front-loading and variability. The first measures the output lost while the government is excluded, the second the value of bringing consumption forward by borrowing, and the third how borrowing changes the volatility of consumption. The gain is the product of the three terms (Appendix A.23). Figure 5 shows the results.

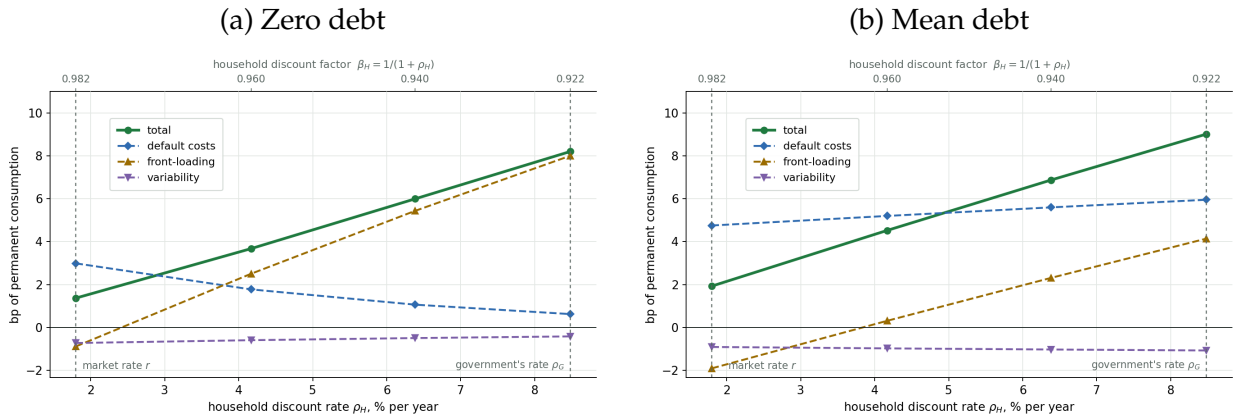


Figure 5. Welfare Gain and the Household's Discount Rate

Notes: Welfare gain from removing the fee, in basis points of permanent consumption, against the household's discount factor, from that of investors, $1/(1+r)$, to that of the government, β. The total is the product of the default-cost, variability and front-loading terms defined in Appendix A.23. Panel (a) evaluates the gain at zero debt and panel (b) at the mean level of debt in the baseline economy.

Households gain at every discount factor between that of the government and that of investors. At zero debt the gain falls from 8.2 basis points for a household as impatient as the government to 1.4 for one as patient as investors (panel a). The two ends differ in where the gain comes from. For the impatient household, front-loading accounts for 8.0 of the 8.2 basis points, and the lower costs of default for 0.6. The patient household gains 3.0 basis points from the lower costs of default and loses 0.9 from the additional borrowing. The variability term is negative at every discount factor, between -0.7 and -0.4 basis points, because the government borrows more and consumption becomes more volatile. At the mean level of debt the lower costs of default are worth between 4.7 and 6.0 basis points, and the gain ranges from 1.9 to 9.0 basis points (panel b).

8 Conclusion

Emerging-market governments borrow both under their own law and under foreign law, usually that of New York or England. If the government defaults, the governing law determines whether domestic or foreign courts settle disputes, and so how much investors recover. This paper studies how these governments borrow under the two laws. We document four facts for twelve governments. First, all twelve borrow under both laws. Second, each issues its foreign-law debt at longer duration than its domestic-law debt. Third, issuing under foreign law costs a fee proportional to the money raised, whatever the maturity of the bond. Fourth, in Brazil, Colombia, Paraguay and Russia the yield gap between domestic- and foreign-law bonds widens with the probability of default.

We develop a quantitative model of sovereign default in which the government borrows under both laws. Through the lens of the model, a yield gap that widens with default risk implies that investors expect to recover more on foreign-law debt in the event of a default. Although foreign-law debt recovers more, the fee leads the government to issue it at longer duration than domestic-law debt, which pays no fee. A short bond must be issued again each time it matures, and each issue pays the fee. The longer duration exposes foreign-law debt to more dilution. We measure the welfare cost of the fee by letting the government issue foreign-law debt at the shorter duration of its domestic-law debt. We find that welfare rises, mostly through changes in how the government borrows and defaults.

Applied to Colombia, the model implies recoveries of 40 percent under domestic law and 47 percent under foreign law. A fee of 25 basis points, Colombia's median, cuts short foreign-law debt from 36 to 3 percent of the government's debt. Issuing foreign-law debt at the duration of its domestic-law debt would raise Colombia's welfare by 8.2 basis points of permanent consumption, 7.4 of which come from changes in the government's borrowing and default policies.

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A Appendix

A.1 Shares of outstanding debt

Inflation-indexed bonds enter the domestic-law stock at their accreted capital: the indexed amount times the current value of the index, which is what the government owes. We convert the domestic-law stock to dollars at the annual average of the year's daily exchange rates. GDP is converted at the same rate, so the exchange rate does not affect debt as a percentage of GDP. The foreign-law share does not involve GDP, so for it the 31 December rate would match the date at which both stocks are measured; Table 12 reports that alternative. A foreign-law tranche counts as foreign law unless its governing-law field in Dealogic names the home country, so a tranche with a blank field falls on the foreign side.

Table 12 reports four alternatives to the baseline share. Columns (2) and (3) convert the domestic-law stock at the 31 December exchange rate and at the annual median of daily rates. Column (4) revalues each foreign-law bond at each year's exchange rate, using the

Table 12. Foreign-Law Share of Marketable Debt: Robustness

	Headline (1)	Dec. FX (2)	Med. FX (3)	Revalued (4)	Net pub. (5)
Brazil	5.0	5.2	5.0	5.1	—
Chile	23.8	24.2	23.7	23.7	23.8
Colombia	29.7	30.2	29.4	29.8	31.6
Hungary	30.2	30.5	30.0	30.2	30.6
Mexico	16.0	16.3	15.9	15.8	—
Paraguay	82.2	82.7	82.2	82.2	—
Peru	34.7	34.8	34.8	34.9	35.5
Poland	20.8	20.8	20.7	20.6	22.2
Russia	20.9	21.6	20.8	20.8	23.5
South Africa	8.5	8.6	8.5	8.7	8.5
Turkey	24.5	26.3	24.5	24.4	28.0
Uruguay	62.5	62.9	62.6	62.4	—

Notes: Column (1) repeats column (4) of Table 1. Paraguay is averaged over 2013–2024, as in Table 1. Columns (2) and (3) convert domestic-law stocks at the 31 December exchange rate and at the annual median of daily rates, instead of the annual average. Column (4) revalues foreign-law bonds at each year's exchange rate instead of keeping their dollar value at issue. Column (5) also removes holdings of public institutions. — : not available.

dual-currency face values that Dealogic records, instead of keeping its dollar value on the issue date. Column (5) also removes the domestic-law holdings of public institutions that bought at competitive auctions. A last exercise, not in the table, removes from the foreign-law side only the tranches with a blank governing-law field whose law cannot be established from another source.

A.2 Inflation-indexed domestic-law debt

Table 13 reports the share of inflation-indexed bonds in the domestic-law marketable debt of each government.

A.3 Duration differences

Each curve in Figure 1 is a Gaussian kernel density estimated separately for one country and one law, over unweighted placements. The bandwidth is 0.45 times the standard deviation of that series, so that the curves of the two laws are smoothed by the same amount relative to their own dispersion, although domestic-law series have far more placements. No correction is applied at zero, so for countries whose domestic-law issuance is domi-

Table 13. Inflation-Indexed Share of Domestic-Law Debt

	Mean, 2006–2024	2024
Brazil	20.4	22.4
Chile	67.2	46.5
Colombia	31.8	33.8
Hungary	0.0	0.0
Mexico	23.5	29.8
Paraguay	0.0	0.0
Peru	7.3	3.1
Poland	1.9	2.1
Russia	2.2	5.6
South Africa	20.3	22.2
Turkey	16.5	10.9
Uruguay	57.3	33.1

Notes: Percent of the domestic-law marketable stock at year-end. Indexed bonds at their indexed value, except Turkey's, at face value. Uruguay from 2010, excluding wage-indexed notes.

nated by short bills the curve understates the mass near zero; the medians and the statistics in the tables are computed from the placements and are not affected.

For bills and zero-coupon bonds, duration equals maturity. For fixed-coupon bonds we apply the formula of Section 2.3 to their coupons and principal, and for inflation-linked bonds we apply it in real terms, with the real coupon and the real yield. For floating-rate bonds, duration equals the time to the next coupon reset. The yield at issuance comes from official sources for domestic-law placements and from Dealogic for foreign-law placements. Duration for a floating-rate bond depends on how we treat its coupon resets, so maturity, which counts every bond over its full term, gives a measure free of that choice; Table 14 reports it, summarized as in Table 2. Table 15 recomputes the medians with every country on 2006–2024, and Table 16 splits foreign-law face by the law under which each placement is written.

Table 14. Maturity at Issuance by Governing Law

	Domestic law				Foreign law				Gap	
	Median (1)	Mean (2)	Face (3)	GDP (4)	Median (5)	Mean (6)	Face (7)	GDP (8)	Median (9)	Face (10)
Brazil	4.59	7.29	4.45	4.36	10.40	16.41	16.24	16.46	5.81	11.79
Chile	9.84	13.81	10.67	10.81	11.42	16.48	16.83	16.60	1.58	6.16
Colombia	4.08	6.64	8.41	—	11.00	17.79	17.76	—	6.92	9.35
Hungary	3.85	4.94	3.08	2.91	8.50	9.39	10.82	10.64	4.65	7.74
Mexico	1.00	4.98	3.31	3.23	10.00	17.25	19.74	19.30	9.00	16.43
Paraguay	3.83	4.92	5.28	4.96	11.41	16.79	16.89	16.94	7.58	11.61
Peru	11.54	15.28	14.05	14.65	16.00	24.20	22.74	22.60	4.46	8.69
Poland	5.26	6.32	5.36	5.15	10.00	11.00	11.34	11.22	4.74	5.98
Russia	7.44	8.33	8.91	8.72	10.00	12.90	14.35	14.31	2.56	5.44
South Africa	9.62	11.34	6.72	6.84	12.00	16.91	17.18	16.90	2.38	10.46
Turkey	2.99	3.89	3.54	3.16	10.00	10.95	10.94	11.37	7.01	7.40
Uruguay	9.53	9.90	10.14	10.00	12.00	16.20	18.92	19.01	2.47	8.78

Notes: Years from settlement to final payment of each placement. Columns as in Table 2. — : not available.

Table 15. Maturity and Duration at Issuance: Common Window

	Maturity		Duration	
	Domestic (1)	Foreign (2)	Domestic (3)	Foreign (4)
Brazil	4.59	10.40	3.47	8.40
Chile	9.84	11.42	8.85	9.90
Colombia	4.08	11.00	3.75	8.71
Hungary	4.29	9.00	2.94	7.43
Mexico	1.00	10.00	0.48	8.87
Paraguay	3.83	11.41	3.46	8.74
Peru	11.54	16.00	8.64	11.77
Poland	5.32	10.00	2.60	8.15
Russia	7.57	10.00	5.62	8.00
South Africa	9.58	12.00	6.89	9.06
Turkey	4.25	10.00	1.64	7.68
Uruguay	9.53	12.00	8.19	9.45

Notes: Median maturity and median duration at issuance, in years, with every country on 2006–2024.

Table 16. Governing Law of Foreign-Law Placements

Country	Governing law (% of foreign-law face)
Brazil	NY 84, London 9, Tokyo 4, Frankfurt 2, Lux 2
Chile	NY 100
Colombia	NY 98, Tokyo 2
Hungary	London 72, NY 21, Tokyo 4, China 2, Frankfurt 1
Mexico	NY 90, Tokyo 8, Frankfurt 2, London 1
Paraguay	NY 100
Peru	NY 100
Poland	London 66, NY 27, Tokyo 5, Frankfurt 1, China 1
Russia	London 96, Frankfurt 4
South Africa	NY 90, London 5, Tokyo 3, Frankfurt 2
Turkey	NY 64, London 23, Frankfurt 7, Tokyo 6
Uruguay	NY 93, Tokyo 7

Notes: Share of foreign-law face issued under each law, from most to least. NY includes tranches that name New York beside the home law.

A.4 A simple model of the yield gap

A government's bonds differ along two dimensions, the law that governs them and the currency they pay, and the two get confounded. Three configurations matter. Two bonds in the same hard currency, differing only in law, give the premium directly; they are rarely issued. A foreign-law bond in dollars against a domestic-law bond in local currency is the usual emerging-market pairing, and the currency conversion it requires fails when default and depreciation coincide. Two bonds in the same local currency, differing only in law, give the premium again, scaled by a factor below one.

Case 1: payoffs in the investor's currency. Suppose a government has two bonds outstanding. Both are zero-coupon bonds maturing next period, both are denominated in dollars, and they are identical except in the law that governs them, $j \in \{D, F\}$. An international investor prices both. She is risk neutral, so she discounts every payoff at the gross risk-free dollar rate $1 + r^{\$}$. The government repays with probability $1 - p$ and defaults with probability p, and it defaults on the two instruments together. After a default a holder of the law-j bond recovers a fraction $\nu_j \in (0, 1)$ of face. Whether foreign law secures the larger recovery, $\nu_F > \nu_D$, is the hypothesis the empirical work tests.

The bond pays one dollar if the government repays and ν_j dollars if it defaults, so

$$q_j^{\$} = \frac{1}{1 + r^{\$}} \Big[(1 - p)\, 1^{\$} + p\, \nu_j\, 1^{\$} \Big], \tag{19}$$

where a superscript \$ marks a quantity denominated in dollars and a superscript LC, used from case 2, one denominated in local currency. Differencing the two prices cancels the common repayment payoff and leaves the recovery difference:

$$\Pi^{\$} \equiv q_F^{\$} - q_D^{\$} = \frac{p\,(\nu_F - \nu_D)\,1^{\$}}{1 + r^{\$}}. \tag{20}$$

The premium is the value of a recovery advantage $\nu_F - \nu_D$ collected only in default. Differentiating with respect to p,

$$\frac{\partial \Pi^{\$}}{\partial p} = \frac{(\nu_F - \nu_D)\,1^{\$}}{1 + r^{\$}}, \tag{21}$$

which is positive under $\nu_F > \nu_D$, so the premium widens with the default probability, and does not respond to it at all under the null $\nu_F = \nu_D$.

Prices are not comparable across bonds of different coupon and maturity, and for several countries in the sample more than one foreign-law bond is outstanding at a time. The premium is therefore measured as the gap between the domestic-law and foreign-law yields. The yield gap carries the sign of the price gap. Since both prices fall as default becomes more likely, it widens with default risk wherever the price gap is positive.

The price premium rises with maturity. The yield premium need not: turning a price gap into a rate per year divides it by the bond's horizon and by its price, and lengthening maturity raises the first and lowers the second, so the direction depends on the parameters. Residual maturity therefore enters the specification as a control.

The model gives the two bonds the same trading costs, but in practice their liquidity differs (Chamon et al., 2018), so the measured gap also reflects differences in the cost of trading. To control for this, the specification includes the foreign-law bond's bid-ask spread relative to that of the country's domestic ten-year benchmark; the domestic-law bond with the same payments is a constructed object and has no quoted spread of its own.

Case 2: currency conversion with a forward. Now suppose instead that the two bonds do not share a currency. This is the usual configuration in emerging markets: the government's foreign-law bonds are denominated in dollars, while its domestic-law bonds pay local currency: 1^{LC} per unit of face under repayment and $\nu_D\,1^{LC}$ in default. Every other feature of case 1 carries over. The same p, ν_D, ν_F, and $r^{\$}$ price both bonds, default still strikes the two instruments together, and the only new object is the exchange rate. Write $\mathcal{E}$ for the spot rate in local currency per dollar, so that a depreciation is a rise in $\mathcal{E}$. When

the bonds mature the rate is $\mathcal{E}'_R$ if the government repays and $\mathcal{E}'_D$ if it defaults, each taken as known conditional on the state. Write $\mathcal{F}$, quoted the same way, for the forward rate contracted today, when the spot rate is $\mathcal{E}$.[11]

The two bonds cannot be compared directly. Converting the local-currency price at today's spot rate does not help: that gives a dollar figure, but the claim still *promises* local currency, so its yield and a dollar yield are not comparable. The comparison requires a claim that promises dollars. Assume a forward on the local currency is available, whether from covered interest parity or from a traded cross-currency swap (Du and Schreger, 2016). Nothing below depends on which.

The forward rate is pinned down by a single no-arbitrage restriction. A unit of local currency is worth $1^{LC}/\mathcal{E}'_R$ dollars under repayment and $1^{LC}/\mathcal{E}'_D$ dollars in default. A forward costs nothing to enter, so it must be worth zero at inception: the $\mathcal{F}$ units of local currency the investor commits to deliver have to be worth, in expectation, exactly the one dollar she receives for them. Under risk-neutral pricing that is

$$\frac{1^{LC}}{\mathcal{F}} = p\left(\frac{1^{LC}}{\mathcal{E}'_D}\right) + (1-p)\left(\frac{1^{LC}}{\mathcal{E}'_R}\right). \tag{22}$$

Write $1-\alpha \equiv \mathcal{F}/\mathcal{E}'_D$, so that α measures the depreciation upon default in excess of what the forward priced.[12] Government defaults coincide with large depreciations, the Twin Ds pattern (Reinhart, 2002; Na et al., 2018). Measured against the no-default state the currency is weaker in default, $\mathcal{E}'_D > \mathcal{E}'_R$ (Mano, 2013), so $\alpha > 0$.

Now build the synthetic foreign-currency domestic-law bond. The investor buys $\mathcal{F}$ units of face of the local-currency bond and sells $\mathcal{F}$ units of local currency forward against one dollar. If the government repays, the bond delivers exactly the $\mathcal{F}$ units of local currency the forward demands, and she collects her dollar whatever the exchange rate turns out to be.

Default is where the construction slips. The forward is written on the face, but the government pays only ν_D of it, so the investor must buy the missing local currency at the post-default rate $\mathcal{E}'_D$ before she can deliver. Her dollar payoff is

$$\left[1-(1-\nu_D)\,\frac{\mathcal{F}}{\mathcal{E}'_D}\right]1^{\$}.$$

[11] $\mathcal{F}$ is a rate, local currency per dollar. As is usual we also treat it as a bare number.

[12] Given $\delta \equiv \mathcal{E}'_D/\mathcal{E}'_R > 1$, the fair-forward condition (22) makes α a function of p, since p prices the forward: $\alpha(p) = 1 - 1/\left[\delta + p\,(1-\delta)\right]$, so $\partial\alpha/\partial p = -\alpha\,(1-\alpha)/(1-p) < 0$. As default becomes more likely the forward is priced closer to $\mathcal{E}'_D$, which is why α falls.

She still collects the dollar in full, because she delivers the whole face either way; that is the first term. The second is what the shortfall costs her. She owes the forward the whole face, the bond delivers ν_D of it, and she buys the remaining $(1-\nu_D)$ at the post-default rate. The whole face would cost $\mathcal{F}/\mathcal{E}'_D$ of a dollar at that rate, so the part she has to buy costs $(1-\nu_D)\mathcal{F}/\mathcal{E}'_D$.

Using $1-\alpha = \mathcal{F}/\mathcal{E}'_D$ and discounting the two payoffs, the bond prices at

$$\widetilde{q}^{\$}_D = \frac{1}{1+r^{\$}}\Big[(1-p)\,1^{\$} + p\,\big(\nu_D + \alpha\,(1-\nu_D)\big)1^{\$}\Big]. \tag{23}$$

The foreign-law bond follows (19), so the measured premium is

$$q^{\$}_F - \widetilde{q}^{\$}_D = \underbrace{\frac{p\,(\nu_F-\nu_D)\,1^{\$}}{1+r^{\$}}}_{\text{legal premium}} - \underbrace{\frac{p\,\alpha\,(1-\nu_D)\,1^{\$}}{1+r^{\$}}}_{\text{quanto}}. \tag{24}$$

The quanto term is the present value of the currency gain $\alpha\,(1-\nu_D)\,1^{\$}$ collected in default. It raises the dollar price of the synthetic bond, so the measured premium falls short of the legal premium in (20), and is itself negative once α exceeds $\alpha^\star \equiv (\nu_F-\nu_D)/(1-\nu_D)$.

Differentiating (24) gives

$$\frac{\partial(q^{\$}_F - \widetilde{q}^{\$}_D)}{\partial p} = \frac{(\nu_F-\nu_D)\,1^{\$}}{1+r^{\$}} - \frac{\bar{\alpha}\,(1-\nu_D)\,1^{\$}}{1+r^{\$}}, \qquad \bar{\alpha} \equiv \alpha + p\,\frac{\partial \alpha}{\partial p}. \tag{25}$$

Because the forward is priced to break even, the α the investor gains in default is offset in expectation by what she loses in repayment. A rise in p makes that gain more likely to be collected, but it also reprices the forward toward $\mathcal{E}'_D$ and so shrinks the gain itself, and $\bar{\alpha}$ combines the two.

Set (25) against (21). The true premium responds to credit risk by $(\nu_F-\nu_D)\,1^{\$}/(1+r^{\$})$, which is positive. The converted premium responds by the same quantity less $\bar{\alpha}\,(1-\nu_D)\,1^{\$}/(1+r^{\$})$, so its slope is negative whenever $\bar{\alpha}$ exceeds $\alpha^\star$.

A back-of-the-envelope calculation at the recoveries in Erce et al. (2024), $\nu_F = 0.67$ on foreign-law bonds against $\nu_D = 0.60$ on domestic-law bonds, gives $\alpha^\star = 0.07/0.40 = 0.175$. Currency-denominated CDS put the depreciation upon default at 36% (Mano, 2013; Du and Schreger, 2016), which fixes $\delta = 1.56$;[13] at $p \to 0$ the $p\,\partial\alpha/\partial p$ term drops out

[13] Du and Schreger (2016) measure the depreciation upon default as the fall in the currency's dollar value. Suppose the rate is 10 pesos per dollar under repayment, so a peso is worth \$0.10. A 36% loss of value

and $\bar{\alpha}$ equals that same 0.36. Substituting into (25) with $r^{\$} = 3\%$, the recovery term is $0.07/1.03 = 0.068$ and the quanto term is $0.36 \times 0.40/1.03 = 0.140$, so the slope is -0.072 per unit of p. A one-point rise in the default probability is $\Delta p = 0.01$, which moves the premium down by 7.2 basis points of face, against 6.8 up from the recovery advantage alone. The slope turns positive again only above a default probability of a third.

The quanto term loads on the same default probability that identifies the premium, so it survives bond fixed effects and first differencing, and a negative estimate is consistent with a substantial legal premium as easily as with none. Removing it requires a design that contracts no forward, which is that of case 3.

Case 3: both bonds in the same currency. The failure documented in case 2 originates in one feature of the construction. The forward rate is contracted today, while the quantity of local currency the investor receives is realized next period. Equation (22) states the consequence: a single rate $\mathcal{F}$ must satisfy one restriction spanning both states, so it cannot equal the state-contingent value of local currency in either, and the discrepancy in the default state is α. That discrepancy is strictly positive under the Twin Ds evidence, and through $\bar{\alpha}$ it enters the estimated slope with the opposite sign to the legal premium. Any design in which a forward is contracted inherits this problem.

We therefore use a comparison in which no forward is contracted at any stage. Several emerging-market governments have issued foreign-law bonds denominated in their own currency, so that a foreign-law and a domestic-law claim on the same government promise identical cash flows in identical units and differ only in the law that governs them. Section 3.3 describes these instruments. The investor is unchanged from case 1, pricing in dollars and discounting at $1 + r^{\$}$; what changes is that the bonds no longer pay in her unit of account.

Take a law-j bond of unit local-currency face. It pays 1 unit of local currency if the government repays, which occurs with probability $1 - p$, and ν_j units of local currency if it defaults, which occurs with probability p. Converting each payoff into dollars at the exchange rate prevailing when it is received, the bond is worth $1^{LC}/\mathcal{E}'_R$ dollars in repayment and $\nu_j 1^{LC}/\mathcal{E}'_D$ dollars in default. Weighting each state by its probability and discounting gives

$$q_j^{\$} = \frac{1}{1 + r^{\$}} \left[(1 - p) \left(\frac{1^{LC}}{\mathcal{E}'_R} \right) + p \left(\frac{\nu_j 1^{LC}}{\mathcal{E}'_D} \right) \right]. \tag{26}$$

leaves it worth \$0.064, at which a dollar buys $1/0.064 = 15.625$ pesos. The rate has gone from 10 to 15.625, so $\delta = 1.56$: the currency lost 36% of its value while the rate rose 56%.

Both bonds promise the same cash flow in repayment, so the repayment term is common to them and cancels in the difference:

$$q_F^{\$} - q_D^{\$} \;=\; \frac{p}{1+r^{\$}}\left(\frac{\nu_F\, 1^{LC}}{\mathcal{E}'_D} - \frac{\nu_D\, 1^{LC}}{\mathcal{E}'_D}\right) \;=\; \frac{p\,(\nu_F - \nu_D)\, 1^{LC}}{(1+r^{\$})\,\mathcal{E}'_D}. \tag{27}$$

Equation (27) is stated per unit of local-currency face, so it carries the scale of the currency the bond is written in and its magnitude is not comparable across governments. We therefore work with prices expressed in local currency. Define

$$\omega \;\equiv\; \frac{\mathcal{E}}{\mathcal{E}'_D},$$

the fraction of its current dollar value that one unit of local currency retains in default. Evaluating (27) at the spot rate,

$$q_F^{LC} - q_D^{LC} \;=\; \mathcal{E}\left(q_F^{\$} - q_D^{\$}\right) \;=\; \frac{p\,(\nu_F - \nu_D)\, 1^{LC}}{1+r^{\$}}\,\omega, \tag{28}$$

$$\frac{\partial\,(q_F^{LC} - q_D^{LC})}{\partial p} \;=\; \frac{(\nu_F - \nu_D)\, 1^{LC}}{1+r^{\$}}\,\omega. \tag{29}$$

The exchange rate enters (29) through ω alone, and $\omega = 1/(g\,\delta)$, where $g \equiv \mathcal{E}'_R/\mathcal{E}$ is the depreciation expected absent default and δ the jump upon default of case 2. It is strictly positive because exchange rates are positive in every state, so the slope has the sign of $\nu_F - \nu_D$ whatever the value of ω. Equation (29) is a difference in prices. The yield gap that we estimate in Section 3.4 carries only the jump at default, $1/\delta$, because the expected depreciation g enters both yields and cancels; Appendix A.5 derives this and measures δ.

The design also has a sharp null. On an instrument sharing the treatment's currency, settlement and clientele but no difference in governing law, $\nu_F = \nu_D$ and (29) is exactly zero for any ω. Conversion returns a strictly negative number instead, so under it a true zero and a masked premium look alike. Appendix A.14 reports this test.

Set (29) against (21) and (25). At the recoveries and depreciation used above, a one-point rise in the default probability moves the true premium of case 1 up 6.8 basis points of face and the converted premium of case 2 down 7.2. At a 10% depreciation absent default, $\omega = 0.58$ and the same-currency premium moves up 4.0.[14] The exchange rate multiplies the same-currency premium and subtracts from the converted one. A positive factor cannot

[14] $\delta = 1.56$ from the 36% of case 2. The slope is 4.3 basis points with no depreciation absent default and 3.5 at 25%.

change a sign, so the same-currency comparison gives the sign of the recovery advantage for any ω, while conversion can report it with the wrong sign. We use the same-currency comparison throughout.

From the model to the data. The model compares two claims with identical cash flows. Actual bonds carry coupons and maturities, no domestic-law bond matches a given foreign-law one, and the currencies at issue are not deliverable offshore. None of this need break the comparison, but each has to be dealt with.

From prices to yields, and maturity. Case 1 measures the premium as a yield gap. Prices there are in dollars throughout. In the one-period case, with $y_j^{\$} = 1/q_j^{\$} - 1$,

$$\Pi^{y} \equiv y_D^{\$} - y_F^{\$} = \frac{1}{q_D^{\$}} - \frac{1}{q_F^{\$}} = \frac{q_F^{\$} - q_D^{\$}}{q_D^{\$}\, q_F^{\$}} = \frac{\Pi^{\$}}{q_D^{\$}\, q_F^{\$}}, \tag{30}$$

so the yield premium is the price premium scaled by the positive factor $1/(q_D^{\$} q_F^{\$})$, and carries its sign. Since $\partial q_j^{\$}/\partial p = (\nu_j - 1)/(1 + r^{\$}) < 0$, both prices fall as p rises, so the factor rises with p and the yield premium widens wherever the price premium does.

The multi-period case. A law-j zero-coupon bond of maturity n and unit face pays 1 at n under repayment and the recovery ν_j at the date of first default otherwise, so under risk-neutral pricing[15]

$$q_j^{(n)} = \sum_{k=1}^{n} (1-p)^{k-1}\, p\, \nu_j\, \frac{1}{(1+r^{\$})^k} \;+\; \frac{(1-p)^n}{(1+r^{\$})^n}. \tag{31}$$

Differencing across the two laws, the common repayment payoff cancels:

$$\Pi^{\$(n)} = p\,(\nu_F - \nu_D)\, A_n(p), \qquad A_n(p) \equiv \sum_{k=1}^{n} \frac{(1-p)^{k-1}}{(1+r^{\$})^k} > 0. \tag{32}$$

The factor $p\, A_n(p) = \sum_{k=1}^{n} p\,(1-p)^{k-1}(1+r^{\$})^{-k}$ is the present value of one unit paid at the first default within n periods, and it rises with p, so $\partial \Pi^{\$(n)}/\partial p > 0$ exactly as in the one-period case (21). Going from n to $n+1$ adds the positive term $(1-p)^n(1+r^{\$})^{-(n+1)}$ to A_n, so A_n rises with n and the premium rises with maturity in price space.

[15] The per-period default hazard p is again common to the two instruments, as in case 1.

Maturity in yield space. Yields are annualized, $y_j^{(n)} = \left(1/q_j^{(n)}\right)^{1/n} - 1$, so the yield premium is

$$\Pi^{y(n)} = \left(q_D^{(n)}\right)^{-1/n} - \left(q_F^{(n)}\right)^{-1/n} \simeq \frac{\Pi^{\$(n)}}{n\left(q^{(n)}\right)^{(n+1)/n}},$$

with $q^{(n)}$ between the two prices. The numerator rises with n, and the divisor carries the two opposing forces: the horizon n over which the yield is annualized, which shrinks the gap, and the price $q^{(n)}$, which falls with n and magnifies it. At $\nu_F = 0.67$, $\nu_D = 0.60$, $r^{\$} = 3\%$ and $p = 0.05$ the yield premium is 37, 38, 38, 35 and 30 basis points at $n = 1, 5, 10, 20, 30$, rising and then falling. The slope also depends on maturity. As the annual hazard rises from one to eight percent, it goes from 0.073 to 0.081 at $n = 1$, from 0.077 to 0.073 at $n = 5$ and from 0.088 to 0.009 at $n = 30$. This is why residual maturity enters the empirical specification (5) only as a control, with no sign restriction.

A.5 Depreciation upon default and the yield-gap estimates

Investors are risk neutral in dollars and discount at the dollar rate r, and the government defaults within the year with probability p. The exchange rate, in local currency per dollar, is $\mathcal{E}$ today, $\mathcal{E}'_R$ next year if the government repays and $\mathcal{E}'_D$ if it defaults. The depreciation upon default is $\delta \equiv \mathcal{E}'_D/\mathcal{E}'_R$, and $\mathcal{E}/\mathcal{E}'_R$ reflects the depreciation expected upon repayment.

The yield gap. A law-j bond pays one unit of local currency if the government repays and ν_j units if it defaults. Its price in local currency is

$$q_j = \frac{\mathcal{E}}{1+r}\left[\frac{1-p}{\mathcal{E}'_R} + \frac{p\,\nu_j}{\mathcal{E}'_D}\right] = \frac{\mathcal{E}}{\mathcal{E}'_R}\,\frac{(1-p) + p\,\nu_j/\delta}{1+r}.$$

With continuously compounded yields, $y_j = -\ln q_j$,

$$y_j = \ln\frac{\mathcal{E}'_R}{\mathcal{E}} + \ln(1+r) - \ln\left[(1-p) + p\,\nu_j/\delta\right].$$

The first two terms are the same for both bonds, so

$$y_D - y_F = \ln\frac{(1-p) + p\,\nu_F/\delta}{(1-p) + p\,\nu_D/\delta} \simeq \frac{p\,(\nu_F - \nu_D)}{\delta}, \tag{33}$$

where the approximation holds for a small probability of default. Over n years the same holds with the annual default probability in place of p, since the jump occurs once.

With annually compounded yields, $y_j = 1/q_j - 1$, the gap is $(q_F - q_D)/(q_D\, q_F) \simeq (1 + i)\, p\, (\nu_F - \nu_D)/\delta$, where $1 + i = (1 + r)\, \mathcal{E}'_R/\mathcal{E}$ is the gross local-currency interest rate. The depreciation expected upon repayment then scales the gap by the level of local-currency rates, the same factor for both bonds.

The CDS ratio. A dollar CDS with one dollar of notional pays $1 - \nu_C$ at default, where ν_C is the recovery set at the ISDA auction, and the buyer pays the spread $s^{\$}$ if the government repays. Its value is zero at inception, so $(1 - p)\, s^{\$} = p\, (1 - \nu_C)$. A local-currency CDS on the same government has $\mathcal{E}$ units of local currency as notional and settles at the same auction and price, so it pays the same fraction $1 - \nu_C$ of its notional, in local currency, against the spread s^{LC}. Valued in dollars, $(1 - p)\, s^{LC}\, \mathcal{E}/\mathcal{E}'_R = p\, (1 - \nu_C)\, \mathcal{E}/\mathcal{E}'_D$. Dividing the first condition by the second,

$$\delta = \frac{s^{\$}}{s^{LC}}. \tag{34}$$

The default probability, the recovery, today's exchange rate and the depreciation expected upon repayment all cancel, and the jump at default remains: the local-currency protection pays in a currency that has lost value, and it is cheaper by that jump. Because δ is a jump at the default date, one value applies to bonds of every maturity.

The slope. With $s^{\$} \simeq p\, (1 - \nu_C)$, the slope of the gap on the dollar CDS is $(\nu_F - \nu_D)/[\delta\, (1 - \nu_C)]$. Multiplying the gap by δ_t removes δ, and the slope of the adjusted gap on the dollar CDS is $(\nu_F - \nu_D)/(1 - \nu_C)$.

Du and Schreger (2016) model depreciation upon default in the same way: the local currency loses a fraction $1 - 1/\delta$ of its dollar value in the default state relative to the non-default state. The depreciation priced at default matches the evidence that defaults coincide with large depreciations (Reinhart, 2002; Na et al., 2018). We compute δ_t daily from Markit's 5-year par spreads, after two screens. The first drops days on which the local-currency spread equals the dollar spread: until a local-currency curve is quoted, Markit republishes the dollar curve under the local-currency ticker. The second drops days with $\delta_t < 1$. Six of the twelve governments in our sample have a quoted local-currency CDS. Table 7 reports δ for five of them over 2015–2025; across them, the local currency loses between 35 and 50 percent of its dollar value at default. Brazil's local-currency CDS is quoted only until May 2015.[16]

[16] Over Brazil's quoted span, 2010–2015, δ averages 1.37. Markit has no local-currency curve for Hungary, Peru and Poland, where the local-currency ticker carries the dollar curve, and none for Chile; we have not obtained local-currency quotes for Paraguay and Uruguay.

Table 17 re-estimates the regressions for Colombia and Russia with the gap multiplied by δ_t day by day. The slopes rise from 0.23 to 0.36 in Colombia and from 0.08 to 0.11 in Russia, as when each slope is multiplied by the mean of δ_t over its window (Table 8).

Table 17. The Yield Gap and Depreciation upon Default

	Gap on dollar CDS		Gap $\times$ δ_t on dollar CDS	
	Slope	s.e.	Slope	s.e.
Colombia	0.227	0.038	0.356	0.060
Russia	0.075	0.030	0.109	0.046

Notes: Each country's first-difference regression of Table 6, without the VIX control, with HC1 standard errors, on days with a screened δ_t. Colombia: 2017–2020, mean δ_t 1.57, 1,373 observations. Russia: 2014–2017, mean δ_t 1.56, 652 observations. Brazil's local-currency CDS covers 15 percent of its window, and Paraguay has none.

A.6 Nonresident access to domestic-law debt

Reading the yield gap as the value of governing law requires that a nonresident can hold the domestic-law bond on the same terms as the foreign-law bond. The foreign-law bond is easy to reach: a nonresident buys it through Euroclear or DTC against dollars. Reaching the domestic-law bond can take more. The investor may have to bring dollars into the country, convert them into local currency, hold the bond through a local custodian, and convert the payments back. Any of these steps can be taxed or restricted, and each one that is raises the yield the investor requires on the domestic-law bond. For each country, we check that capital can enter and leave without controls or entry taxes, and that the currency trades at a single onshore rate open to nonresidents. Each sample starts only after these conditions hold.

Brazil. Brazil meets the conditions from June 2013. Until then, a nonresident who bought domestic-law bonds had to convert dollars into reais onshore and paid the IOF tax on the conversion: 2 percent on portfolio inflows from October 2009, and 4 and then 6 percent on fixed-income inflows in October 2010.[17] The tax applied to the onshore purchase and not to the Global BRL bonds, which are bought and settled in dollars. It was cut to zero on 4 June 2013.[18] Brazil's sample starts in January 2014, the first full year after the tax was removed.

[17] Decreto 6.983 de 19 de outubro de 2009; Decreto 7.323 de 4 de outubro de 2010; Decreto 7.330 de 18 de outubro de 2010, each amending Decreto 6.306 de 2007.

[18] Decreto 8.023 de 4 de junho de 2013, amending art. 15-A, incisos XI and XII, of Decreto 6.306 de 2007.

Colombia. Colombia meets the conditions throughout its sample. The unremunerated reserve requirement on portfolio inflows, imposed in May 2007 at 40 percent and raised to 50 percent in May 2008, was abolished in October 2008.[19] Foreign portfolio investors have a legal right to remit profits and the proceeds of sales abroad in convertible currency.[20] The one difference between the two bonds is a withholding tax on the domestic-law bond (Appendix A.7). Colombia's sample starts in June 2017 for a different reason: that month the longest of its foreign-law bonds falls within ten years of maturity, the range of the domestic-law curve (Appendix A.10).

Paraguay. Paraguay meets the conditions from January 2025. On 26 December 2024, the central bank revised its rules on issuance, custody and trading and opened the domestic market to nonresidents.[21] Before then, a nonresident had no practical way to hold the domestic-law bonds. Funds can be converted and repatriated without restriction, and the guaraní floats.[22] Paraguay's sample starts on 6 February 2025, when Bloomberg's prices for the domestic-law bonds begin (Appendix A.10).

Russia. Russia meets the conditions from February 2013, when Euroclear began settling its domestic-law government bonds (Lu and Yakovlev, 2017). Before then, a nonresident needed a local custodian. The rouble floated freely from November 2014.[23] The sanctions adopted in July 2014 barred new debt of named banks and energy firms, but left the government's own bonds and their secondary trading untouched.[24] Russia's sample runs from March 2014 to March 2017, while its foreign-law bond has between four years and one year left. The panel has no observations from 22 December 2014 to 27 April 2015, because quotes on the ten-year domestic benchmark used for the liquidity control are missing.

Peru. Peru meets the conditions throughout, which makes it usable as the placebo of Section 3.4. The constitution guarantees the free holding and disposal of foreign currency,

[19] Decreto 1801 de 2007; Decreto 1888 de 2008; abolished by Decreto 3913 de 8 de octubre de 2008.

[20] Decreto 1068 de 2015, arts. 2.17.2.2.3.1 and 2.17.2.2.3.2, as modified by Decreto 119 de 2017.

[21] Banco Central del Paraguay, Directorio, Resolución N.º 25, Acta N.º 62 de 26 de diciembre de 2024, which admits global custodian banks and qualified institutional investors and opens secondary trading to foreign holders.

[22] IMF, *Paraguay: 2024 Article IV Consultation*, Staff Report; U.S. Department of State, *Investment Climate Statement: Paraguay*.

[23] IMF, *Russian Federation: 2015 Article IV Consultation*, Country Report No. 15/211.

[24] OFAC Directive 1 under E.O. 13662, 16 July 2014, as amended 12 September 2014; Council Regulation (EU) No 833/2014 of 31 July 2014.

the sol floats, and no controls apply to portfolio flows.[25] Bonos Soberanos have settled through Euroclear since 2017.[26]

Holdings. Table 18 reports the share of each government's domestic-law marketable debt held by nonresidents, from national sources: the central bank, the treasury or the debt office. The first two columns are means over the full period of each source, and the last two are means over the estimation window of the yield gap (Table 5). The columns marked "all holders" divide nonresident holdings by the debt held by every holder, including the central bank and public institutions. The columns marked "net of public" remove the holdings of the central bank and of public institutions and funds from the denominator, which leaves the debt that trades among private investors.

Table 18. Nonresident Holdings of Domestic-Law Debt

		Full period		Estimation window	
	Source, period	All holders	Net of public	All holders	Net of public
Brazil	Tesouro Nacional, 2007–2026	11.4	12.0	15.2	16.1
Chile	Banco Central, 2012–2026	9.5	—		
Colombia	MinHacienda, 2010–2026	17.3	20.0	25.6	29.2
Hungary	ÁKK, 2004–2026	29.1	30.7		
Mexico	Banxico, 2004–2026	20.9	21.8		
Paraguay	MEF, 2025–2026	3.5	—	—	—
Peru	MEF, 2007–2026	41.4	42.6	43.6	45.2
Poland	Ministry of Finance, 2004–2026	25.4	27.5		
Russia	Bank of Russia, 2011–2026	19.1	24.3	22.9	32.7
South Africa	National Treasury, 2006–2026	31.6	43.7		
Turkey	Treasury, 2004–2026	12.2	16.6		
Uruguay	Debt office, 2024–2026	2.2	—		

Notes: Percent of domestic-law marketable government debt held by nonresidents, monthly or quarterly means. Net of public removes: Brazil, government funds (the central bank is already excluded); Colombia, the central bank and the core public sector; Hungary, Poland and Turkey, the central bank and the government sector or public banks; Mexico, Banxico's holdings; Peru, public funds; Russia, the Bank of Russia and direct-placement tranches; South Africa, the government pension fund. Paraguay: two observations; Uruguay: implied from the debt office's composition reports. Estimation windows: Brazil 2014–2020, Colombia 2017–2020, Russia 2014–2017, Peru 2017–2025 excluding March 2020 to July 2021.

A.7 Taxes on the two bonds

A withholding tax on the domestic-law bond that the foreign-law bond escapes would enter the yield gap. Colombia is the only one of the five countries where nonresidents pay

[25] Constitución Política del Perú (1993), art. 64; U.S. Department of State, *Investment Climate Statement: Peru*.

[26] Reglamento de Bonos Soberanos, D.S. N.º 309-2016-EF; Resolución Ministerial N.º 059-2017-EF/52.

such a tax. Brazil has exempted nonresidents' income from federal government bonds since 2006.[27] Russia paid coupons on its domestic-law bonds to nonresidents gross until 2021, after its sample ends.[28] Paraguay exempts nonresidents' income from public debt.[29] Peru exempts interest on government bonds for residents and nonresidents alike.[30] The foreign-law bonds pay gross in all five countries.

In Colombia, nonresident portfolio investors paid a 14 percent withholding tax on income from domestic-law bonds from 2013, reduced to 5 percent from 2019.[31] The Global TES pay gross: Colombia undertakes to pay additional amounts so that holders receive their payments free of any Colombian withholding (Republic of Colombia, 2007, clause 5(a)).

The tax is levied on income, so a nonresident earns approximately $(1 - m_t)\,\widetilde{y}_{D,t}$ on the domestic-law bond, where m_t is the tax rate. The tax adds a wedge $m_t\,\widetilde{y}_{D,t}$ to the yield gap. The wedge is proportional to the domestic-law yield, which rises with default risk, so first differences do not remove it. We net the tax out by measuring the gap as $(1 - m_t)\,\widetilde{y}_{D,t} - y_{F,t}$, with m_t equal to 14 percent before 2019 and 5 percent from 2019. This moves Colombia's slope from 0.225 to 0.212 with the VIX control, and from 0.221 to 0.186 without it (Table 23). The correction applies the full statutory rate. Some holders bear less, through tax treaties or by crediting the tax at home, so the correction is an upper bound on the effect of the tax.

A.8 Settlement of the foreign-law bonds

Each foreign-law bond promises fixed amounts of local currency. Brazil's, Colombia's and Paraguay's pay those amounts in dollars, because the real, the Colombian peso and the guaraní cannot be delivered through DTC, Euroclear or Clearstream. Paying in dollars changes the currency in which the bond settles, not the currency of the claim. Russia's bond is the exception. The rouble could be delivered through Euroclear, so the bond pays

[27] Lei 11.312 de 27 de junho de 2006, art. 1, for bonds acquired from 16 February 2006 under the rules of the Conselho Monetário Nacional, excluding investors domiciled in jurisdictions that tax income at less than 20 percent.

[28] Tax Code of the Russian Federation, art. 310, which relieves the payer of withholding on interest paid to foreign organizations on state securities. The standard rate on these bonds for nonresidents was zero until 31 December 2020.

[29] Ley 6380/2019, art. 79, numeral 4.

[30] Texto Único Ordenado de la Ley del Impuesto a la Renta, D.S. N.º 179-2004-EF, art. 19, inciso h). The exemption carries a sunset, most recently extended to 31 December 2026.

[31] Ley 1607 de 2012, art. 125, amending Estatuto Tributario art. 18-1, effective from tax year 2013. The reduction to 5 percent was enacted by Ley 1943 de 2018, art. 72, and applied from 2019. That law was declared unconstitutional by Sentencia C-481 de 2019 with effect from 1 January 2020, and the same provision was re-enacted by Ley 2010 de 2019. Investors domiciled in listed jurisdictions face 25 percent.

roubles unless the holder chooses dollars, and we use its Euroclear-settled tranche.

The conversion uses the onshore reference rate, computed from transactions inside the country and published by the central bank or the financial regulator: the PTAX in Brazil and the Representative Market Rate in Colombia and Paraguay.[32] Each is a spot rate set shortly before the payment, so no forward and no cross-currency basis enters the bond. It is also the rate at which a holder of the domestic-law bond converts the same payments into dollars. An offshore rate or a non-deliverable forward rate would instead open a wedge between the two bonds.

A.9 Defaults on both classes of bonds

The model assumes that the government defaults on both classes of bonds at once. We assess this assumption with the database of Erce et al. (2024), January 2026 release, which records government default and restructuring events on domestic and external debt. Our model concerns marketable bonds held by investors who can buy bonds under both laws, so we select the episodes that match this setting in three steps. Table 19 reports the events in the database, Table 20 the episodes after the first two steps, and Table 21 the episodes after the third.

First, we keep bonds that private investors bought in the market. The database records 297 events in 84 countries, of which 184 are bank loans and commercial credit and 113 are bonds. Using the authors' companion volume of sovereign histories, which names the instrument in each episode, we keep the 90 bond events in which an instrument traded in the secondary market. Grouping these events into episodes as the authors do gives 43 episodes in 31 countries.

Second, we keep episodes in which bonds under both laws were outstanding. We measure the existence of foreign-law bonds with Dealogic, as bonds issued before the episode and maturing after it, and the existence of a domestic bond market from the authors' episode narratives. Bonds under both laws were outstanding in 27 of the 43 episodes. In 12 of them, both classes of bonds were restructured; in 8, only domestic-law bonds; and in 7, only foreign-law bonds.

Third, among the 15 episodes in which only one class was restructured, we keep those in which foreign investors could hold the domestic-law debt. In 11 of them, exchange

[32] Colombia's rate is a five-business-day average of the weighted average of buy and sell rates on transactions by banks in Bogotá, Cali, Barranquilla and Medellín, published by the Superintendencia Financiera (Republic of Colombia, 2007). Brazil's is the commercial rate the central bank reports on the SISBACEN system. Paraguay's is fixed by the Banco Central del Paraguay ten business days before each payment.

controls or terms that applied only to residents limited the domestic-law debt to resident holders. Among the remaining episodes, both classes of bonds were restructured in 12 of 14 under a strict classification and in 12 of 16 under a lenient one.

Table 19. Default Events in the Database

	Events	Percent	US$ billion
All events	297	100.0	1,671.6
Bank loans and commercial credit	184	62.0	736.9
Bonds	113	38.0	934.8
	Domestic	Foreign	Total
Bond events: bought by private investors	59	31	90
Bond events: other instruments	23	0	23

Notes: Events in Erce et al. (2024), January 2026 release. A bond event counts as bought by private investors if any instrument in it was priced in the secondary market. Domestic and foreign refer to the market of issuance.

Table 20. Default Episodes on Bonds Bought by Private Investors

	All episodes		Bonds under both laws outstanding	
	Episodes	Percent	Episodes	Percent
Both classes hit	12	27.9	12	44.4
Domestic-law bonds only	18	41.9	8	29.6
Foreign-law bonds only	13	30.2	7	25.9
Total	43	100.0	27	100.0

Notes: Episodes as published by the authors. The episodes that hit both classes are Argentina 2001, Barbados 2018, Côte d'Ivoire 2011, Dominica 2003, Ecuador 1997, Greece 2011, Grenada 2004 and 2013, Russia 1998, St Kitts and Nevis 2011, Ukraine 1998 and Uruguay 2002.

Table 21. Episodes in Which Foreign Investors Could Hold Domestic-Law Debt

	Strict		Lenient	
	Episodes	Percent	Episodes	Percent
Both classes hit	12	85.7	12	75.0
One class restructured	2	14.3	4	25.0
Total	14	100.0	16	100.0

Notes: Of the 15 episodes in which only one class was restructured, 11 are dropped because the domestic-law debt was limited to resident holders: Belize 2006, 2012 and 2016, Brazil 1986, Cyprus 2013, Jamaica 2010 and 2013, Mozambique 2015 and 2016, Seychelles 2008 and Venezuela 1998. Strict keeps Turkey 1999 and Panama 1998; lenient also keeps Venezuela 2002 and Pakistan 1998.

A.10 Instruments and curves

Table 22 lists the bonds in the yield-gap sample. Two short bonds never enter: Brazil's 12.5% '16 and Colombia's 12% '15.

Table 22. Bonds in the Yield-Gap Sample

Country	Bond	ISIN	Governing law
Brazil	Global BRL 12.5% '22	US105756BL31	New York
	Global BRL 10.25% '28	US105756BN96	New York
	Global BRL 8.5% '24	US105756BT66	New York
Colombia	Global TES 9.85% '27	XS0306322065	New York
	Global TES 7.75% '21	XS0502258444	New York
	Global TES 4.375% '23	XS0833886095	New York
Paraguay	Global 7.9% '31	US699149AP51	New York
	Global 8.5% '35	US699149BX76	New York
	Global 8.5% '38	US699149CC21	New York
Russia	7.85% '18	XS0564087541	English
Peru (placebo)	GDN 6.90% '37	US715638AV48	Peruvian[a]
	GDN 6.95% '31	US715638BE14	Peruvian[a]
	GDN 6.35% '28	USP78024AC31	Peruvian[a]

[a] New York law governs the deposit agreement; the claim on the government is the underlying Bono Soberano and remains under Peruvian law.
Notes: Regulation S tranches unless stated. Paraguay's 7.9% '31 has a Regulation S twin, USP75744AP07; Russia's Rule 144A tranche is US783064AN80.

For Brazil and Colombia we use Bloomberg's domestic-law zero curves, and for Russia

the central bank's. Brazilian rates are quoted as effective annual rates on a business-day clock of 252 days a year, and we price Brazil's bonds on the same convention. We use Colombia's curve over maturities of one to ten years, so each Colombian bond enters the sample once its remaining maturity falls within ten years. The 2027 bond enters in June 2017, which sets the start of Colombia's sample. Brazil's and Colombia's samples end at the COVID break in February 2020, Russia's when its bond falls below one year to maturity, and Paraguay's at the end of the available data.

Paraguay has no published zero curve for its domestic-law guaraní bonds, so we fit one each day. We use the eleven domestic-law Bonos del Tesoro that Bloomberg prices with BVAL, its evaluated price; between 9 and 11 of them are quoted on a given day. The bonds pay semiannual coupons, and we compute accrued interest on a 30/360 basis. Bloomberg's BVAL history for these bonds begins on 6 February 2025, which sets the start of Paraguay's sample.

On each day we fit the Nelson–Siegel curve

$$y(t) = \beta_0 + \beta_1 \frac{1 - e^{-t/\lambda}}{t/\lambda} + \beta_2 \left(\frac{1 - e^{-t/\lambda}}{t/\lambda} - e^{-t/\lambda} \right),$$

where $y(t)$ is the annually compounded zero yield at a horizon of t years. Given the curve, we price each bond by discounting each remaining cash flow at the zero yield of its own date, $P^{\text{model}} = \sum_k CF_k \left(1 + y(t_k)\right)^{-t_k}$. We choose $(\beta_0, \beta_1, \beta_2)$ to minimize

$$\sum_{\text{bonds}} \left(\frac{P^{\text{model}} - P^{\text{market}}}{\text{DV01}} \right)^2,$$

where P^{market} is the bond's dirty mid price and DV01 is its price change for a one-basis-point change in its yield. Dividing by DV01 turns each price error into a yield error, so long bonds do not dominate the fit. We drop bonds within 30 days of maturity and days with fewer than four quotes.

The decay parameter λ sets where the curvature term peaks. Each day we select λ from a grid between 0.15 and 3 years, and we use the 21-day rolling median of the daily choices. Of the alternatives we compared, this one keeps the fit within 25 percent of the best daily fit and moves the curve least from one day to the next. λ equals 3 on 278 of the 331 days.

We use the curve only within the maturities quoted that day, from the shortest remaining domestic bond to the longest. A foreign-law bond enters the sample on a given day only

if its final payment falls within that range. Over the 331 days, the median root-mean-squared yield error of the fit is 7 basis points. We also refit the curve without one bond at a time and price the omitted bond on it. The median absolute error is 7 basis points when the omitted bond lies inside the range of the others and 16 basis points when it lies at either end; this is why we do not extrapolate. Fitting the curve separately to bid and to ask prices moves it by about 8 basis points at ten years and 12 at five years, measured as half the distance between the two curves.

Figure 6 shows the fit on four dates. Each red point is a domestic-law bond's yield to maturity, plotted at its duration. For a coupon bond this yield is close to the zero yield at its duration, so the vertical distance from a point to the line approximates that bond's fitting error. The crosses place each foreign-law Global at its own duration on the fitted curve. The range of quoted maturities and the number of bonds change across dates as domestic bonds mature and new ones are issued, and the 2038 Global enters in February 2026.

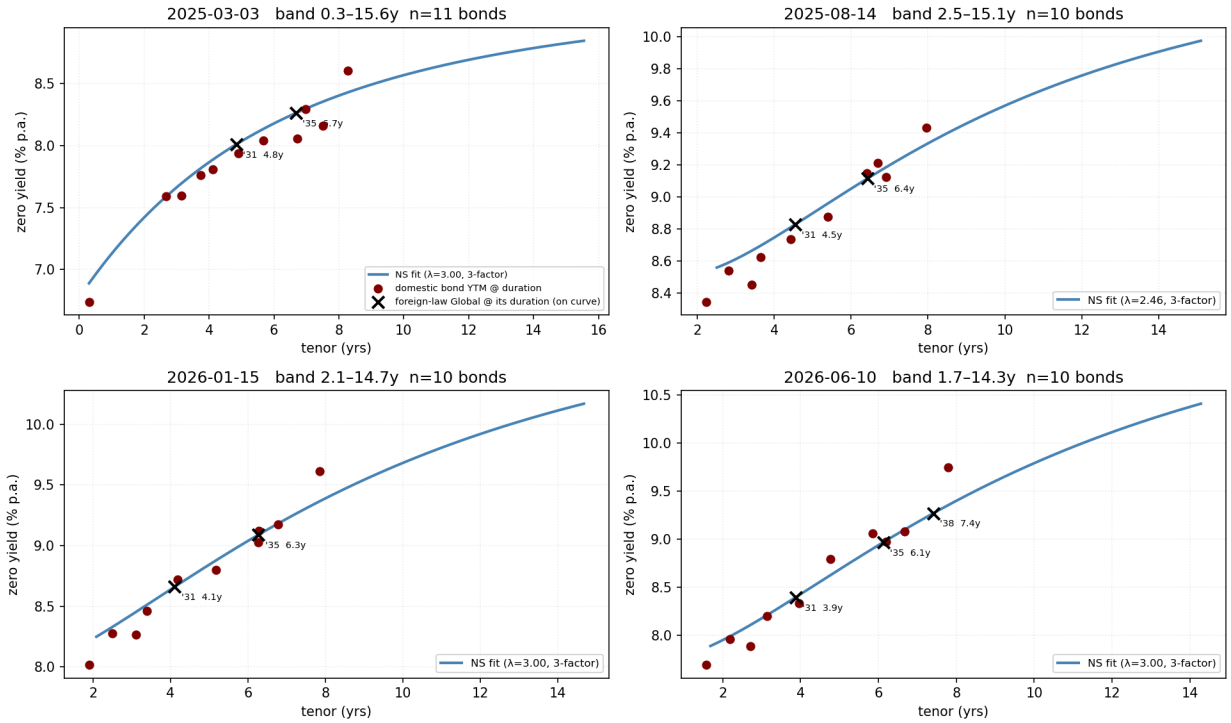


Figure 6. Paraguay's Fitted Domestic-Law Curve

Notes: Daily Nelson–Siegel zero curve fitted to the domestic-law Bonos del Tesoro on four dates. Points: each domestic-law bond's yield to maturity at its duration. Crosses: each foreign-law Global at its duration, on the fitted curve. Panel titles give the range of quoted maturities and the number of domestic-law bonds in the fit.

A.11 Stationarity, cointegration and inference

Both series behave as near-random walks. Augmented Dickey–Fuller tests cannot reject a unit root in the level of the yield gap or of the CDS spread in Brazil, Colombia, Paraguay and Peru. KPSS tests reject stationarity in the same series, and the first differences are stationary on both tests. Russia is the exception: there the yield gap is stationary while the CDS spread is not, so a regression in levels would be unbalanced for a different reason.

Cointegration would justify a regression in levels, and it is mostly absent. The Engle–Granger test rejects the null of no cointegration only for one bond in Brazil ($p = 0.019$ for the 10.25% '28) and one in Colombia ($p = 0.012$ for the 9.85% '27), and only over the period before COVID. The rejection disappears once the later data are added, and it never appears for Paraguay or Peru.

Time to maturity enters equation (5) in levels, as in Chamon et al. (2018). Differencing it turns it into a near-deterministic day counter and leaves the slope unchanged.

The Driscoll–Kraay standard errors use the automatic Newey–West bandwidth $L = \lfloor 4\,(T/100)^{2/9} \rfloor$ over the T trading days. The correction widens the confidence bands where daily changes are positively autocorrelated, in Brazil and Colombia, and narrows them where they are negatively autocorrelated, in Paraguay and Russia.

A.12 Robustness of the yield-gap estimates

Table 23 re-estimates equation (5) under alternative specifications, country by country, without and with the change in the VIX. The first rows report the Driscoll–Kraay t-statistic of the baseline, drop the liquidity control, drop the maturity control, and keep only differences between consecutive trading days. The window rows extend Brazil's sample past the COVID break, from July 2020 or from July 2021 to December 2025; extend Colombia's to 2017–2025 and to 2015–2025, excluding the COVID period; exclude the Russian crisis months from November 2014 to March 2015; and extend Peru's sample to 2017–2025 including the COVID period. The last rows drop one bond at a time. Two further rows interact the change in the relative bid-ask spread with the level of the CDS spread, so that the effect of liquidity can vary with default risk, and, for Colombia, measure the gap net of the withholding tax that nonresidents paid on domestic-law bonds.

Table 23. The Yield Gap and the CDS Spread: Robustness

	Without ΔVIX		With ΔVIX	
	Slope	t	Slope	t
Brazil				
Baseline	0.669^{***}	(10.65)	0.694^{***}	(9.72)
Driscoll–Kraay t-statistic	0.669^{***}	(7.37)	0.694^{***}	(6.80)
Without the liquidity control	0.668^{***}	(10.62)	0.694^{***}	(9.68)
Without the maturity control	0.669^{***}	(10.66)	0.694^{***}	(9.74)
One-trading-day differences	0.703^{***}	(8.96)	0.738^{***}	(8.21)
Liquidity control × CDS level	0.666^{***}	(10.57)	0.692^{***}	(9.69)
Adding July 2020 to December 2025	0.590^{***}	(10.83)	0.631^{***}	(10.21)
Adding July 2021 to December 2025	0.628^{***}	(11.28)	0.676^{***}	(10.64)
Without the 2022 bond	0.657^{***}	(7.97)	0.686^{***}	(7.27)
Without the 2028 bond	0.681^{***}	(10.29)	0.706^{***}	(9.42)
Without the 2024 bond	0.659^{***}	(7.56)	0.682^{***}	(6.87)
Colombia				
Baseline	0.221^{***}	(5.74)	0.225^{***}	(4.83)
Driscoll–Kraay t-statistic	0.221^{***}	(4.35)	0.225^{***}	(3.81)
Without the liquidity control	0.221^{***}	(5.74)	0.225^{***}	(4.83)
Without the maturity control	0.221^{***}	(5.74)	0.225^{***}	(4.83)
One-trading-day differences	0.104^{***}	(3.16)	0.117^{***}	(2.93)
Liquidity control × CDS level	0.218^{***}	(5.73)	0.220^{***}	(4.82)
Net of the withholding tax	0.186^{***}	(5.16)	0.212^{***}	(4.56)
2017–2025, excluding the COVID period	0.231^{***}	(3.25)	0.250^{***}	(2.90)
2015–2025, excluding the COVID period	0.187^{***}	(3.47)	0.203^{***}	(3.10)
Without the 2027 bond	0.265^{***}	(5.25)	0.260^{***}	(4.32)
Without the 2021 bond	0.205^{***}	(4.55)	0.229^{***}	(4.00)
Without the 2023 bond	0.193^{***}	(4.31)	0.188^{***}	(3.55)
Paraguay				
Baseline	0.157^{**}	(2.03)	0.169^{**}	(2.10)
Driscoll–Kraay t-statistic	0.157^{**}	(2.20)	0.169^{**}	(2.16)
Without the liquidity control	0.222^{***}	(2.83)	0.231^{***}	(2.81)
Without the maturity control	0.158^{**}	(2.03)	0.170^{**}	(2.10)
One-trading-day differences	0.109	(0.96)	0.108	(0.90)
Liquidity control × CDS level	0.153^{**}	(1.98)	0.164^{**}	(2.07)
Without the 2031 bond	0.298^{***}	(2.94)	0.364^{***}	(3.43)
Without the 2035 bond	0.088	(0.87)	0.059	(0.58)
Without the 2038 bond	0.117	(1.49)	0.123	(1.54)
Russia				
Baseline	0.079^{***}	(2.64)	0.073^{**}	(2.30)
Driscoll–Kraay t-statistic	0.079^{***}	(3.38)	0.073^{***}	(2.83)
Without the liquidity control	0.098^{***}	(2.80)	0.097^{**}	(2.54)
Without the maturity control	0.079^{***}	(2.64)	0.072^{**}	(2.30)
One-trading-day differences	0.028	(0.76)	0.018	(0.46)
Liquidity control × CDS level	0.062^{**}	(2.13)	0.054^{*}	(1.79)
Excluding November 2014 to March 2015	0.059^{*}	(1.75)	0.049	(1.38)
Peru (placebo)				
Baseline	−0.028	(−1.46)	−0.017	(−0.76)
Driscoll–Kraay t-statistic	−0.028	(−1.08)	−0.017	(−0.58)
Without the liquidity control	−0.027	(−1.42)	−0.018	(−0.78)
Without the maturity control	−0.028	(−1.46)	−0.017	(−0.75)
One-trading-day differences	-0.038^{*}	(−1.93)	-0.046^{**}	(−2.05)
Liquidity control × CDS level	−0.028	(−1.46)	−0.017	(−0.76)
2017–2025, including the COVID period	−0.021	(−1.01)	−0.022	(−0.95)
Without the 2037 bond	−0.014	(−0.54)	0.011	(0.33)
Without the 2031 bond	-0.054^{***}	(−2.87)	-0.052^{**}	(−2.41)
Without the 2028 bond	−0.017	(−0.72)	−0.012	(−0.41)

Notes: Estimates of γ in equation (5), without and with the change in the VIX among the controls, on the days for which the VIX is available. t-statistics in parentheses use Huber–White standard errors, except in the Driscoll–Kraay row (Newey–West for Russia, which has one bond). Stars follow the t-statistic shown: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.13 Binned scatters by country and bond-by-bond estimates

A.14 Placebo: Peru's Global Depositary Notes

Peru's sol Global Depositary Notes give a placebo for the yield gap. Each note is a claim, issued by a depositary bank, on domestic-law Bonos Soberanos held in custody in Peru. New York law governs the deposit agreement, and the payment claim remains under Peruvian law, so both sides of the gap are domestic-law claims. Nonresidents can buy, hold and sell the Bonos Soberanos without restrictions, and held 40.9 percent of their outstanding stock at the end of 2024.[33]

We use the notes on the Bonos Soberanos maturing in 2028, 2031 and 2037, from April 2017 to December 2025, excluding March 2020 to July 2021, and apply the construction and the regression of Section 3.3. The slope of the gap on Peru's CDS spread is -0.017, with a Driscoll–Kraay t-statistic of -0.58 and 3,391 observations. Table 23 reports the alternative specifications.

A.15 The cost of issuing

We estimate equation (1), with the fee in basis points of the money raised and size in US$ billion raised. The sample is the New York-law bonds with a fee, without two retail notes that Mexico sold in 2006, and without the bonds below the 1st and above the 99th percentile of maturity: three bonds shorter than 5 years and four longer than 50 years. This leaves 360 bonds from ten countries. Standard errors are clustered by country, and because there are only ten clusters we also report p-values from a wild bootstrap. Neither coefficient is significant: $p = 0.41$ for maturity (0.51 with the bootstrap) and $p = 0.24$ for size (0.27), and the joint test that both are zero gives $p = 0.23$. With ten clusters, the clustered covariance matrix cannot test the eighteen year effects jointly, so the joint tests of the country and of the year effects in Table 4 use heteroskedasticity-robust errors. With the seven bonds outside the percentiles included, the coefficient on maturity is 0.048, with $p = 0.25$. For Colombia, the same regression on its 49 New York-law bonds, with year effects and errors clustered by deal, gives $p = 0.63$ for maturity and $p = 0.63$ for size (Figure 7).

Only five countries have bonds with a recorded fee under laws other than New York law, 50 bonds in all, and three quarters of them are yen bonds issued under Japanese law. Table 26 compares the fee by law and market. English-law bonds in euros pay about the

[33] Ministerio de Economía y Finanzas, *Informe Anual de Deuda Pública 2024* (May 2025).

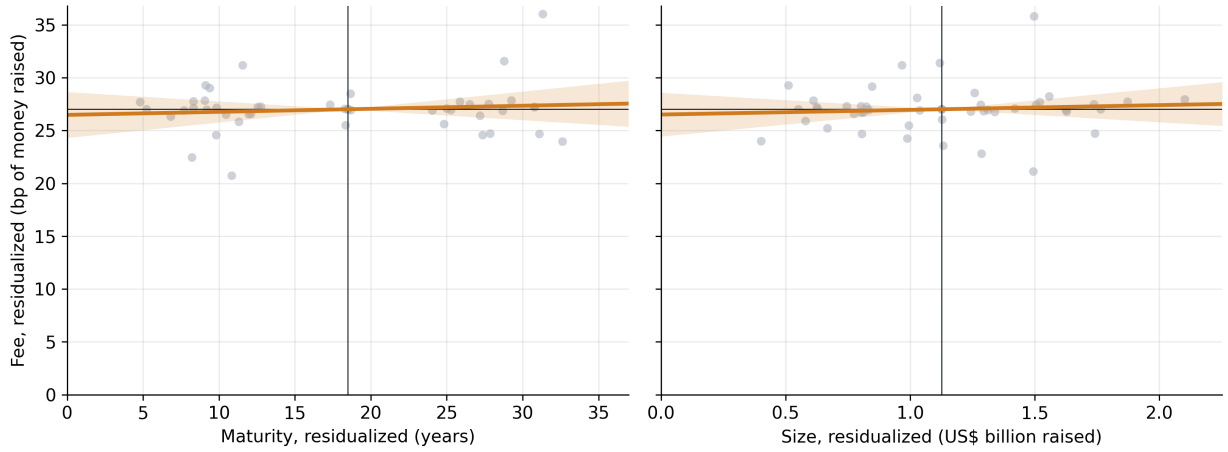


Figure 7. Underwriting Fee, Maturity and Size: Colombia

Notes: As Figure 2, for Colombia's 49 New York-law bonds, 2006–2024, with year effects; averages added back (fee 27.0 basis points, maturity 18.5 years, size US$1.1 billion). Shaded: 95 percent interval of the slope, with errors clustered by deal.

PROSPECTUS SUPPLEMENT
(TO PROSPECTUS DATED APRIL 15, 2021)

U.S. $2,000,000,000

Republic of Colombia

3.250% Global Bonds due 2032

The bonds will mature on April 22, 2032. The Republic of Colombia ("Colombia" or the "Republic") will pay interest on the bonds each April 22 and October 22, commencing on October 22, 2021. The bonds will be issued in denominations of U.S. $200,000 and integral multiples of U.S. $1,000 in excess thereof.

The bonds will be direct, general, unconditional, unsecured and unsubordinated external indebtedness of Colombia and will be backed by the full faith and credit of Colombia. The bonds will rank without any preference among themselves and equally with all other unsecured and unsubordinated external indebtedness of Colombia. It is understood that this provision shall not be construed so as to require Colombia to make payments under the bonds ratably with payments being made under any other external indebtedness.

Colombia may, at its option, redeem the bonds, in whole or in part, before maturity, on not less than 10 nor more than 60 days' notice on the terms described under "Description of the Bonds—Optional Redemption" in this prospectus supplement. The bonds will not be entitled to the benefit of any sinking fund.

The bonds will be issued under an indenture and constitute a separate series of debt securities under the indenture. The indenture contains provisions regarding future modifications to the terms of the bonds that differ from those applicable to Colombia's outstanding public external indebtedness issued prior to January 28, 2015. Under these provisions, which are described beginning on page 7 of the accompanying prospectus, Colombia may amend the payment provisions of any series of debt securities (including the bonds) and other reserve matters listed in the indenture with the consent of the holders of: (1) with respect to a single series of debt securities, more than 75% of the aggregate principal amount of the outstanding debt securities of such series; (2) with respect to two or more series of debt securities, if certain "uniformly applicable" requirements are met, more than 75% of the aggregate principal amount of the outstanding debt securities of all series affected by the proposed modification, taken in the aggregate; or (3) with respect to two or more series of debt securities, more than 66 2/3% of the aggregate principal amount of the outstanding bonds of all series affected by the proposed modification, taken in the aggregate, and more than 50% of the aggregate principal amount of the outstanding debt securities of each series affected by the proposed modification, taken individually.

Application will be made to list the bonds on the official list of the Luxembourg Stock Exchange and to trade them on the Euro MTF Market of the Luxembourg Stock Exchange.

See "Risk Factors" beginning on page S-7 to read about certain risks you should consider before investing in the bonds.

Section 309B(1)(c) of the Securities and Futures Act (Chapter 289 of Singapore) Notification

The bonds are prescribed capital markets products (as defined in the Securities and Futures (Capital Markets Products) Regulations 2018).

Neither the Securities and Exchange Commission, referred to as the SEC, nor any other regulatory body has approved or disapproved of these securities or determined if this prospectus supplement or the accompanying prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

	Per bond	Total
Public offering price(1)	99.032%	U.S. $1,980,640,000
Underwriting discount	0.200%	U.S. $ 4,000,000
Proceeds, before expenses, to Colombia	98.832%	U.S. $1,976,640,000

(1) Purchasers will also be required to pay accrued interest, if any, from April 22, 2021, if settlement occurs after that date.

Source: Republic of Colombia, prospectus supplement (424B5) for the U.S.$2,000,000,000 3.250% Global Bonds due 2032, April 2021. SEC EDGAR accession 0001193125-21-124194, cover page. Highlight added.

Figure 8. The Cover of a Prospectus

Table 24. Issuer Expenses per Deal

Country	Prospectuses with an expense figure (1)	Expenses, median, US$ thousand (2)	Expenses, s.d., US$ thousand (3)	Share of the cost of a deal, % (4)
Brazil	29	185	121	5
Chile	30	150	106	20
Colombia	34	300	44	7
Hungary	7	175	71	5
Mexico	40	250	88	4
Peru	14	265	73	17
Poland	11	200	79	6
South Africa	14	305	187	12
Turkey	1	—	—	—
Uruguay	20	120	40	10

Notes: Expenses the issuer pays outside the underwriting fee (legal, listing, agents), as estimated in each SEC prospectus, New York-law bonds, 2006–2024. Where the stated figure includes the fee, the fee is subtracted. Column (4) is the median, deal by deal, of expenses over the fee plus expenses.

Table 25. Underwriting Fee on Bonds under Other Foreign Laws

	Bonds (1)	Median (2)	Mean (3)	Mean, face-weighted (4)
Hungary	6	30.0	33.3	22.0
Mexico	18	26.0	26.6	25.8
Poland	21	35.0	38.8	35.3
Turkey	1	13.0	13.0	13.0
Uruguay	4	8.0	8.0	8.0

Notes: As in Table 3, for bonds issued under Japanese, English and Swiss law, 2006–2024.

Table 26. Underwriting Fee by Governing Law and Market

	All countries			Same country: median		
Law and market	Bonds	Median	Mean	Mexico	Poland	Hungary
New York law	368	15.0	16.5	19.0	14.0	18.0
English law, euro	8	14.0	14.2	—	14.5	10.0
Japanese law, yen	37	29.0	29.2	26.0	35.0	30.0
Swiss franc	4	75.0	81.2	—	75.0	75.0
English law, hand-collected	37	14.0	13.5	—	14.0	12.5

Notes: Underwriting fee in basis points of face value, Dealogic, 2006–2024. The last three columns give the median fee of each country's bonds in each market. The last row uses English-law fees read from Luxembourg and London exchange filings.

same fee as New York-law bonds, 14 against 15 basis points, and in Poland 14.5 against 14.0. Yen bonds pay about twice as much, and Swiss-franc bonds more. Among these 50 bonds, a 30-year bond pays 5.5 basis points less than a 10-year bond of the same country and year, with a 95 percent interval from -8.8 to -2.1.

A.16 How often and how much governments issue

Governments borrow abroad less often than at home, and in larger operations. Table 27 reports, for each country, how often the government issues under each law and how much it raises each time, by quarter, over 2006–2024. The share of quarters with no foreign-law issuance ranges from 0.26 in Turkey to 0.83 in Paraguay. Under domestic law it is at most 0.25, and six countries issue domestic-law bonds every quarter. Relative to that quarter's GDP, the median foreign-law operation is larger than the median domestic-law one in ten of the twelve countries, with a ratio that ranges from 1.1 in Poland to 90 in Uruguay; Brazil and Turkey are the two exceptions.

The unit of count is the operation: one day on which bonds settle in a market, with everything that settles that day added up. Re-openings of outstanding foreign-law bonds count as operations, because each is a new sale abroad with its own prospectus and fee. Both laws use the placements of the previous sections: Dealogic for foreign law and debt-office records for domestic law. We convert each operation to US dollars and divide it by that quarter's nominal GDP in US dollars.

Table 27. Issuance by Governing Law, Quarterly

	Inaction rate		Operations per quarter		Spacing,	One operation, % of quarterly GDP		
	Foreign (1)	Domestic (2)	Foreign (3)	Domestic (4)	quarters (5)	Foreign (6)	Domestic (7)	Ratio (8)
Brazil	0.54	0.00	0.61	20.1	2.1	0.22	0.48	0.5
Chile	0.75	0.17	0.36	3.4	3.1	2.51	0.45	5.5
Colombia	0.55	0.09	0.50	8.4	2.3	1.95	0.28	7.0
Hungary	0.62	0.00	0.51	10.0	2.6	3.38	0.58	5.9
Mexico	0.34	0.00	0.84	15.1	1.5	0.61	0.25	2.4
Paraguay	0.83	0.25	0.17	1.7	3.7	5.11	0.15	34.6
Peru	0.80	0.05	0.22	6.7	4.4	2.54	0.09	26.8
Poland	0.41	0.00	0.99	5.4	1.7	0.92	0.81	1.1
Russia	0.80	0.01	0.20	8.6	3.1	0.61	0.09	7.0
South Africa	0.80	0.00	0.21	19.5	5.3	1.87	0.19	9.7
Turkey	0.26	0.00	1.12	5.6	1.4	0.73	0.76	1.0
Uruguay	0.64	0.19	0.39	4.0	2.9	7.95	0.09	90.0

Notes: 2006Q1–2024Q4. An operation is one settlement day in a market, with amounts summed. Foreign-law placements from Dealogic, counting re-openings as operations; domestic-law placements from debt-office records, bonds only. Exchange operations are excluded on both sides. Columns (1)–(2) give the share of quarters with no issuance, (3)–(4) the operations per quarter, (5) the mean number of quarters between foreign-law issuance quarters, (6)–(7) the median operation as a percentage of that quarter's GDP, and (8) the ratio of (6) to (7).

A.17 Recovered debt in the numerical solution

We solve the model on a uniform grid for each bond. Domestic-law debt takes the values $b_D(i) = \bar{b}_D\, i/(N_D - 1)$ for $i \in \{0, 1, \ldots, N_D - 1\}$, and foreign-law debt is defined in the same way on N_F points. A government that regains access after a default carries $\nu_D b_D$ and $\nu_F b_F$. Because each grid is linear in its index and starts at zero, scaling the debt level by the recovery is the same as scaling the index: $\nu_D b_D(i) = b_D(\nu_D i)$. The recovered position $p_D = \nu_D i$ is generally not an integer, and the same holds for $p_F = \nu_F j$. Since recoveries are at most one, the recovered point never leaves the grid.

Each recovered position lies between two grid points, $i^- = \lfloor p_D \rfloor$ and $i^+ = \min(i^- + 1, N_D - 1)$, with weight $a = p_D - i^-$, and likewise j^-, j^+ and $c = p_F - j^-$ for foreign-law debt. We evaluate the new-issuance prices q_D^{ND} and q_F^{ND} and the continuation value at the recovered point by bilinear interpolation of the four surrounding grid values,

$$\tilde{f}(p_D, p_F) = (1-a)(1-c)\, f[i^-, j^-] + a(1-c)\, f[i^+, j^-] + (1-a)c\, f[i^-, j^+] + ac\, f[i^+, j^+]. \tag{35}$$

The weights are nonnegative and sum to one, and when a recovered position falls on a grid point the formula uses that point alone. Each law's recovery therefore scales its own

debt, and the recovered portfolio need not lie on the grid.

A.18 The two experiments

Table 28. Same Durations, Different Recoveries: Foreign-Law Share (%)

	Duration of both bonds (years)				
ν_D / ν_F	1.00	2.00	3.75	5.50	8.71
0.200 / 0.400	100.0	100.0	100.0	100.0	100.0
0.400 / 0.420	100.0	100.0	100.0	100.0	100.0
0.400 / 0.471	100.0	100.0	100.0	100.0	100.0
0.400 / 0.550	100.0	100.0	100.0	100.0	100.0
0.400 / 0.700	100.0	100.0	100.0	100.0	100.0
0.471 / 0.400	0.0	0.0	0.0	0.0	0.0

Notes: Foreign-law share of debt when both bonds have the same duration. Other parameters at their calibrated values. The last row reverses which law has the higher recovery.

Table 29. Same Durations, Different Recoveries: Recovery Grid (%)

	ν_F				
ν_D	0.300	0.375	0.450	0.525	0.600
0.300	39.7	100.0	100.0	100.0	100.0
0.375	0.0	39.7	100.0	100.0	100.0
0.450	0.0	0.0	45.6	100.0	100.0
0.525	0.0	0.0	0.0	43.7	100.0
0.600	0.0	0.0	0.0	0.0	39.1

Notes: Foreign-law share of debt, both bonds with a duration of 3.75 years. On the diagonal the two bonds are identical and the split is not determined by the model.

Table 30. Same Recoveries, Different Durations: Foreign-Law Share (%)

	Duration, domestic / foreign law (years)		
$\nu_D = \nu_F$	1 / 8.71	2 / 8.71	3.75 / 8.71
0.00	47.8	2.0	0.3
0.40	58.8	2.0	0.3
0.4715	60.4	2.0	0.1
0.60	65.3	2.0	0.1
0.75	67.8	3.7	0.1

Notes: Foreign-law share of debt when both bonds have the same recovery. The last column uses the calibrated durations. Other parameters at their calibrated values.

A.19 Proof of Proposition 1

Events. Fix the government's policies and start at date 0 from a state $s_0 = (y_0, b_{D,0}, b_{F,0})$ at which one unit of each bond is issued. Along a path of output and reentry draws, the policies determine in each period $t \geq 1$ whether the government repays and whether it reenters:

$$\chi_t = \begin{cases} 1 & \text{repays at } t \\ 0 & \text{otherwise} \end{cases}, \qquad \eta_t = \begin{cases} 1 & \text{reenters at } t \\ 0 & \text{otherwise} \end{cases}, \qquad \chi_t \, \eta_t = 0.$$

The government reenters only from default, so it never repays and reenters in the same period. The events are the same for both bonds: the path of states follows from output and the policies, the government repays both bonds or defaults on both, and it reenters on all its debt at once.

Holdings and payments. Let $h_{j,t}$ be the units of law-j debt held at the start of t, before the events of t, by an investor who bought one unit at date 0. At a repayment a fraction λ matures, and at reentry each unit becomes ν_j units:

$$h_{j,1} = 1, \qquad h_{j,t+1} = h_{j,t} \times \begin{cases} 1 - \lambda & \chi_t = 1, \\ \nu_j & \eta_t = 1, \\ 1 & \text{otherwise.} \end{cases}$$

With $a_t \equiv \sum_{s<t} \chi_s$ repayments and $k_t \equiv \sum_{s<t} \eta_s$ reentries before t,

$$h_{j,t} = (1-\lambda)^{a_t}\, \nu_j^{k_t}, \qquad c_{j,t} = \chi_t\, \lambda\, h_{j,t} = \chi_t\, \lambda\, (1-\lambda)^{a_t}\, \nu_j^{k_t},$$

where $c_{j,t}$ is the payment at t: the matured fraction of the holdings if the government repays, and zero otherwise.

Prices as present values. Investors are risk neutral, so the price of a bond is the expected present value of its payments:

$$q_j^{ND}(s_0) = \mathbb{E}_0 \sum_{t=1}^{\infty} \frac{c_{j,t}}{(1+r)^t}. \tag{36}$$

Equations (11) and (12) are the recursive form of (36). The same expression gives $q_j^D(s_0)$, with the path starting in default at date 0 and reentry possible at date 0.

Comparison. By (36),

$$q_F^{ND}(s_0) - q_D^{ND}(s_0) = \mathbb{E}_0 \sum_{t=1}^{\infty} \frac{\chi_t\, \lambda\, (1-\lambda)^{a_t}}{(1+r)^t} \left(\nu_F^{k_t} - \nu_D^{k_t}\right) \;\geq\; 0,$$

since $\nu_F \geq \nu_D \geq 0$ gives $\nu_F^k \geq \nu_D^k$ for every $k \geq 0$. The same holds for $q_F^D - q_D^D$.

Strictness. If $\nu_F > \nu_D$, then $\nu_F^k > \nu_D^k$ for every $k \geq 1$, so each term with $\chi_t = 1$ and $k_t \geq 1$ is positive. The gap is positive whenever, with positive probability, the government reenters and then repays. If the government never defaults from s_0 then $k_t = 0$ in every period and $q_F^{ND}(s_0) = q_D^{ND}(s_0)$.

A.20 The fee and the cost of borrowing

This appendix shows why a fee charged on the money raised at each issue weighs more on short bonds than on long ones.

What the fee is charged on. The fee f is a share of the money the government raises, and it is paid once, when the bond is issued (Section 2.4).

The bond. A bond with decay λ and no coupon pays, per unit of face, λ next period, $\lambda(1-\lambda)$ the period after, and so on: each period a share λ of the face still outstanding

matures. Its yield y is the constant rate that discounts these payments to the bond's price q:

$$q = \sum_{k=1}^{\infty} \frac{\lambda(1-\lambda)^{k-1}}{(1+y)^k} = \frac{\lambda}{1+y} \sum_{k=0}^{\infty} \left(\frac{1-\lambda}{1+y}\right)^k = \frac{\lambda}{1+y} \frac{1+y}{y+\lambda} = \frac{\lambda}{y+\lambda}. \tag{37}$$

Solving for the yield gives $y = \lambda/q - \lambda$. A one-period bond has $\lambda = 1$ and $y = 1/q - 1$. For a given yield, the price rises with λ: a bond that repays sooner is discounted less, so short bonds trade closer to par than long bonds.

The fee as a lower price. With the fee, the government sells one unit of face for q but keeps only $(1-f)\,q$, while it owes the same payments. The yield it pays on the money it actually receives is therefore

$$\hat{y} = \frac{\lambda}{(1-f)\,q} - \lambda. \tag{38}$$

Subtracting y and using $q = \lambda/(y+\lambda)$ from equation (37),

$$\hat{y} - y = \frac{\lambda}{q} \frac{f}{1-f} = (y+\lambda) \frac{f}{1-f}. \tag{39}$$

Short against long bonds. We compare the wedge on two bonds of the same government that differ only in their decay rate and trade at the same yield. At a given yield, the wedge in equation (39) rises one for one with λ, so it is larger on the shorter bond.

The table evaluates these expressions at the fee of 25 basis points and a common yield of 4.2 percent, the risk-free rate plus Colombia's mean spread. The first two rows are the short and the long foreign-law bonds of Section 6.2; the third is the foreign-law bond of Section 5.1. To keep one unit of face outstanding, the government issues λ units each period and pays $fq\lambda$ in fees; the last column is the wedge in the yield.

Bond	λ	Price q	$fq\lambda$ (bp of face)	Wedge $(y+\lambda)f/(1-f)$ (bp a year)
One period	1.000	0.960	24.0	26.1
2.32 years	0.420	0.909	9.5	11.6
8.71 years	0.099	0.702	1.7	3.5

From one period to 8.71 years, the wedge falls by a factor of seven. For the wedge on a long bond to reach that on the one-period bond, its yield would have to exceed the one-period yield by the difference in λ: by 90 percentage points for the 8.71-year bond and by 58 for the 2.32-year bond.

A.21 Proof of Proposition 2

This appendix sets out the full environment of the simple model of Section 6.1, states Proposition 2 in full and proves it.

Environment. Time is discrete and infinite, $t = 0, 1, 2, \ldots$, and there is no uncertainty. The government receives a constant endowment y. It values consumption at $\sum_{t \geq 0} \beta^t u(c_t)$, with u strictly increasing, strictly concave and differentiable. Lenders discount at the interest rate r, and $\beta = 1/(1+r)$. The government always repays. It starts period 0 owing $B_0 > 0$, due in full in period 0.

At $t = 0$ the government chooses a decay rate $\xi \in [0, 1/\beta)$, and every bond it issues, at that date and later, has this decay rate. A unit issued at t pays 1 at $t+1$, ξ at $t+2$, ξ^2 at $t+3$, and so on. Lenders price it at $q = \beta/(1-\beta\xi)$.

Let i_t be the units issued at t, with $i_t < 0$ for a buyback, and b_t the payment due at t on the bonds issued from period 0 on, so that b_{t+1} is the debt the government chooses at t. The government pays a fee $f \in [0, 1)$ on the market value of what it issues or buys back. Its budget constraints are

$$c_0 = y - B_0 + q\, i_0 - f\, q\, |i_0|, \qquad c_t = y - b_t + q\, i_t - f\, q\, |i_t| \quad (t \geq 1), \tag{40}$$

with $b_1 = i_0$ and $b_{t+1} = \xi b_t + i_t$ for $t \geq 1$, and its debt is bounded, $b_{t+1} \leq \bar{b}$ for all $t \geq 0$, with $\bar{b} \geq \max\{B_0,\ (1-\beta)\, B_0/((1-f)\,\beta)\}$. A plan $\{c_t, i_t\}_{t \geq 0}$ is feasible under maturity ξ if it satisfies (40) in every period and the debt limit.

Proposition A.1. *(i) If $f = 0$, every ξ gives the same allocation, $c_t = y - (1-\beta)\, B_0$ for all t.*

(ii) If $f \in (0, 1)$ and $(1-f)\, u'(y - B_0) > u'(y)$, the government borrows at $t = 0$ and the consol, $\xi = 1$, is the unique optimal maturity. The government issues only at $t = 0$. Consumption is c_0^* at $t = 0$ and $\bar{c}^*$ at every $t \geq 1$, where

$$(1-f)\, u'(c_0^*) = u'(\bar{c}^*), \qquad c_0^* < \bar{c}^*.$$

(iii) If $f \in (0, 1)$ and $(1-f)\, u'(y - B_0) \leq u'(y)$, the government does not borrow, $c_0 = y - B_0$ and $c_t = y$ for $t \geq 1$, and every ξ gives this allocation.

Proposition 2 in Section 6.1 summarizes parts (i) and (ii).

Proof. Fix ξ and write $q = q(\xi)$.

Step 1: intertemporal budget constraint. Iterating the law of motion, $b_t = \sum_{s<t} \xi^{t-1-s} i_s$. A unit issued at s is worth $\beta^s q$ at date 0. Its payments up to T are worth $\sum_{t=s+1}^{T} \beta^t \xi^{t-1-s}$, and those after T are worth $\xi^{T-s}\beta^T q$. Summing over all issues up to T,

$$\sum_{s=0}^{T} \beta^s q\, i_s = \sum_{t=1}^{T} \beta^t b_t + \beta^T q\, b_{T+1}. \tag{41}$$

Multiplying each constraint in (40) by β^t, adding up to T and using (41),

$$\sum_{t=0}^{T} \beta^t c_t = y\,\frac{1-\beta^{T+1}}{1-\beta} - B_0 + \beta^T q\, b_{T+1} - f\, q \sum_{t=0}^{T} \beta^t |i_t|.$$

By the debt limit, $\beta^T q\, b_{T+1} \le \beta^T q\, \bar{b} \to 0$. Letting $T \to \infty$,

$$\sum_{t\ge 0} \beta^t c_t \le \frac{y}{1-\beta} - B_0 - f\, q \sum_{t\ge 0} \beta^t |i_t|. \tag{42}$$

Step 2: a constraint common to every maturity. Let $x = c_0 - (y - B_0)$ be the net amount raised at $t = 0$. By (40), $x = q\, i_0 - f\, q\, |i_0|$, so x has the sign of i_0, and the fee paid at $t = 0$ is $f\, q\, |i_0| = \varphi(x) - x$, with

$$\varphi(x) = \begin{cases} x/(1-f), & x \ge 0, \\ x/(1+f), & x < 0. \end{cases}$$

Separating $t = 0$ in (42) and dropping the fees paid after $t = 0$, every feasible plan satisfies, whatever ξ,

$$\sum_{t\ge 1} \beta^t c_t \le \frac{\beta\, y}{1-\beta} - \varphi(x). \tag{43}$$

The fee at $t = 0$ does not depend on ξ: every bond pays it.

Step 3: the relaxed problem. Maximize $\sum_{t\ge 0} \beta^t u(c_t)$ subject to (43) alone. Every plan that is feasible under some maturity satisfies (43). The value of the relaxed problem is therefore an upper bound on the value of any feasible plan, and a feasible plan that reaches it is optimal.

Given c_0, the right side of (43) is fixed. Because $\beta(1+r) = 1$, consumption after $t = 0$ is constant at $\bar{c}$, and the constraint binds:

$$\bar{c}(c_0) = y - \frac{1-\beta}{\beta}\,\varphi\big(c_0 - (y - B_0)\big).$$

The problem reduces to maximizing $W(c_0) = u(c_0) + \frac{\beta}{1-\beta}\, u\big(\bar{c}(c_0)\big)$. The function φ is convex, with slope $1/(1+f)$ below zero and $1/(1-f)$ above, so $\bar{c}$ is concave in c_0 and W is strictly concave. Its maximizer is unique. Away from the kink at $c_0 = y - B_0$,

$$W'(c_0) = u'(c_0) - \varphi'(x)\, u'\big(\bar{c}(c_0)\big).$$

At the kink $\bar{c} = y$. The left derivative is $u'(y - B_0) - u'(y)/(1+f) > 0$, since $B_0 > 0$, so saving is never optimal. The right derivative is $u'(y - B_0) - u'(y)/(1-f)$. If $(1-f)\, u'(y - B_0) > u'(y)$, the right derivative is positive and the maximizer c_0^* satisfies $(1-f)\, u'(c_0^*) = u'(\bar{c}^*)$, with $\bar{c}^* = \bar{c}(c_0^*)$. Then $u'(c_0^*) > u'(\bar{c}^*)$, so $c_0^* < \bar{c}^*$, and the government raises $x^* = c_0^* - (y - B_0) > 0$. Otherwise the right derivative is not positive, and the maximizer is the kink: $c_0^* = y - B_0$ and $\bar{c}^* = y$.

Step 4: the consol attains the relaxed optimum. Let $\xi = 1$, so $q = \beta/(1-\beta)$. Issue $i_0 = (1-\beta)\, x^*/((1-f)\,\beta)$ at $t = 0$ and nothing afterwards. Then $b_t = i_0$ and $c_t = y - i_0$ for every $t \geq 1$, and

$$y - i_0 = y - \frac{1-\beta}{\beta}\,\frac{x^*}{1-f} = \bar{c}^*.$$

No fee is paid after $t = 0$ and $\beta^T q\, i_0 \to 0$, so (42) and (43) hold with equality. Since $c_0^* < \bar{c}^* < y$, we have $x^* < B_0$, so $i_0 < (1-\beta)\, B_0/((1-f)\,\beta) \leq \bar{b}$ and the debt limit holds.

Step 5: no other maturity attains it. Suppose $f > 0$, the government borrows, and a plan with $\xi \neq 1$ attains the relaxed optimum. Then (43) holds with equality, so no fee is paid after $t = 0$: $i_t = 0$ for $t \geq 1$. Hence $b_t = \xi^{t-1} i_0$ and $c_t = y - \xi^{t-1} i_0$, with $i_0 > 0$ because $x^* > 0$. Then $c_2 - c_1 = (1-\xi)\, i_0 \neq 0$, so consumption after $t = 0$ is not constant, which contradicts Step 3. This proves part (ii).

Step 6: the remaining cases. In part (iii), $x^* = 0$, and issuing nothing at any date gives $c_0 = y - B_0$ and $c_t = y$ for every ξ. In part (i), $\varphi(x) = x$ and (43) becomes $\sum_{t\geq 0} \beta^t c_t \leq y/(1-\beta) - B_0$, whose solution is $c_t = y - (1-\beta)\, B_0$ for all t. Any ξ attains it by keeping the market value of debt at $q\, b_{t+1} = \beta B_0$ in every period. Then $c_0 = y - B_0 + q\, b_1 = y - (1-\beta)\, B_0$. For $t \geq 1$, $q\, i_t = q\, b_{t+1} - \xi\, q\, b_t = (1-\xi)\,\beta B_0$ and

$$c_t = y - \frac{\beta B_0}{q} + (1-\xi)\,\beta B_0 = y - (1-\beta)\, B_0.$$

Face value is $b_{t+1} = (1 - \beta\xi)\, B_0 \leq B_0 \leq \bar{b}$. □

A.22 The fee ladder

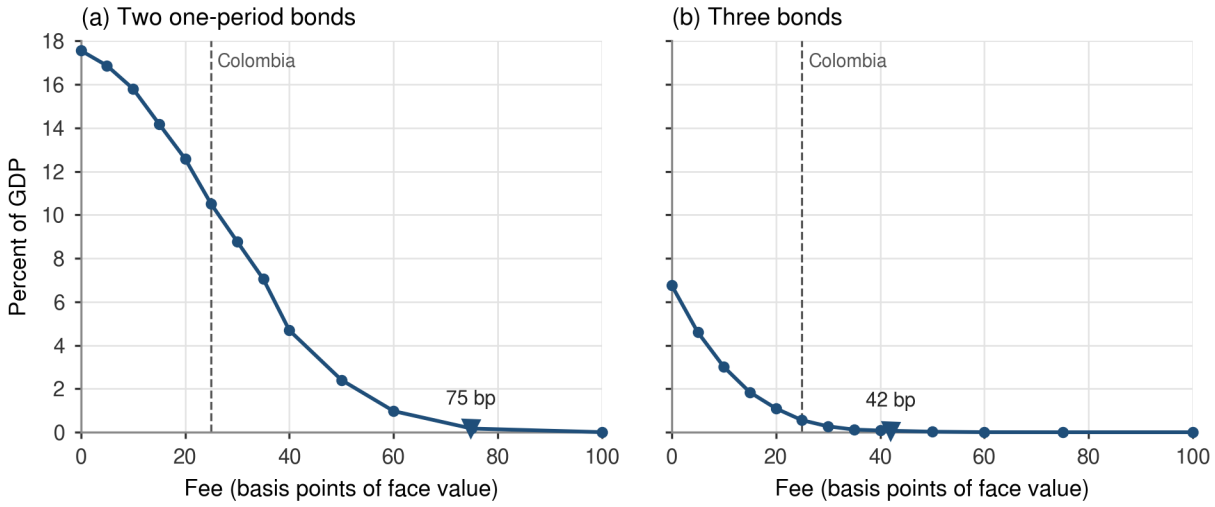


Figure 9. Foreign-Law Debt and the Fee

Notes: Mean debt in percent of GDP against the fee on the money raised under foreign law. Panel (a): foreign-law debt when the government has a one-period domestic-law bond and a one-period foreign-law bond. Panel (b): short foreign-law debt when the government also has a foreign-law bond with a duration of 2.32 years. The dashed line marks Colombia's fee of 25 basis points. The triangle marks the fee at which the plotted debt falls below 1 percent of its value without a fee.

Table 31. Shares of Debt and the Foreign-Law Fee

	Two bonds	Three bonds		
Fee (bp of money raised)	Foreign law	Short domestic	Short foreign	Long foreign
0	100.0	0.0	35.9	64.1
20	75.7	11.2	5.7	83.1
25 (Colombia)	63.0	13.3	2.9	83.8
35	43.9	17.1	0.8	82.0
Fee at which issuance stops (bp)				
Foreign law, two bonds	100			
Short foreign law, three bonds			35	

Notes: Shares of total face value, in percent, means over simulated periods. Two bonds: a one-period domestic-law bond and a one-period foreign-law bond. Three bonds: one-period domestic-law and foreign-law bonds and a foreign-law bond with a duration of 2.32 years. The fee is charged on the money raised by issues and buybacks of foreign-law bonds. Issuance stops when the share falls below 1 percent, among the fees we solve. Other parameters as in Table 9.

A.23 Welfare: measurement and decomposition

Decomposition with policies fixed. The economy without the fee is the baseline with the short bond issued under foreign law, so that it recovers 0.47. The price of defaulted debt,

$$q_i^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\,\mathbb{E}_{y'|y}\left[q_i^D(y', b_D, b_F)\right] + \theta\,\nu_i\,q_i^{ND}(y, \nu_D b_D, \nu_F b_F)\,,$$

shows the two roles of the recovery: a higher ν_i raises the price investors pay, and it raises the debt the government carries at reentry. Holding the borrowing and default policies at their baseline values, we compute welfare in four economies: the baseline; new prices with the baseline debt at reentry; baseline prices with the new debt at reentry; and both changes together. The first comparison isolates the price effect and the second the reentry effect. The difference between the economy with $\nu_D = 0.47$, in which policies adjust, and the economy with both changes and fixed policies is the effect of the government's behavior.

The value of market access. For an economy $\mathcal{E}$, let $\bar{V}^{\mathcal{E}}$ be the household's expected discounted utility along the equilibrium path, with consumption $y - \phi(y)$ while the government is excluded. Let $\bar{V}^{ND,\mathcal{E}}$ be the same value when exclusion carries no output loss, so that consumption equals y while excluded and the government still reenters owing $\nu_D b_D$ and $\nu_F b_F$. Let $\bar{V}(\bar{c}^{ND,\mathcal{E}}) = \sum_t \beta_H^t u(\bar{c}_t^{ND,\mathcal{E}})$ be the value of receiving for sure the mean path $\bar{c}_t^{ND,\mathcal{E}}$ of that consumption. Let $\bar{V}^A$ be the value of consuming output forever, and $\bar{V}^A(y^\infty)$ the value of consuming mean output $y^\infty = \sum_y \pi(y)\,y$ forever. Following Aguiar et al. (2020), the consumption-equivalent value of market access is

$$1 + \chi^{\mathcal{E}} \equiv \left(\frac{\bar{V}^{\mathcal{E}}}{\bar{V}^A}\right)^{\frac{1}{1-\sigma}} = \underbrace{\left(\frac{\bar{V}^{\mathcal{E}}}{\bar{V}^{ND,\mathcal{E}}}\right)^{\frac{1}{1-\sigma}}}_{\text{default costs}} \underbrace{\left(\frac{\bar{V}^{ND,\mathcal{E}}}{\bar{V}(\bar{c}^{ND,\mathcal{E}})}\,\frac{\bar{V}^A(y^\infty)}{\bar{V}^A}\right)^{\frac{1}{1-\sigma}}}_{\text{variability}} \underbrace{\left(\frac{\bar{V}(\bar{c}^{ND,\mathcal{E}})}{\bar{V}^A(y^\infty)}\right)^{\frac{1}{1-\sigma}}}_{\text{front-loading}}. \tag{44}$$

The value of autarky is the same in the two economies, so the welfare gain of equation (18) is $1 + \Delta = (1 + \chi^{\mathcal{E}_1})/(1 + \chi^{\mathcal{E}_0})$, and each term contributes the ratio of its own values across the two economies.